AGT1 - MODULE-1

Objectives

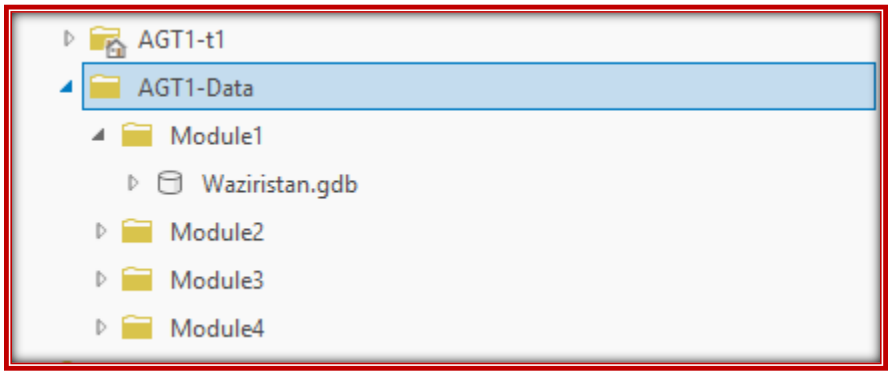
- Generate contours.
- Generate hillshade.
- Calculate slope.
- Calculate aspect.
- Calculate viewshed.

Deliverables (To be submitted)

- Map showing contours with index contours highlighted using **both** methods (Workspace screenshot not layout)
- Map showing hillshade for Waziristan DEM (Workspace screenshot not layout) Map showing slope raster for Waziristan DEM (Workspace screenshot not layout)
- Map showing aspect raster for Waziristan DEM (Workspace screenshot not layout)
- Map showing viewshed for given tower locations using Waziristan DEM (Workspace screenshot not layout)

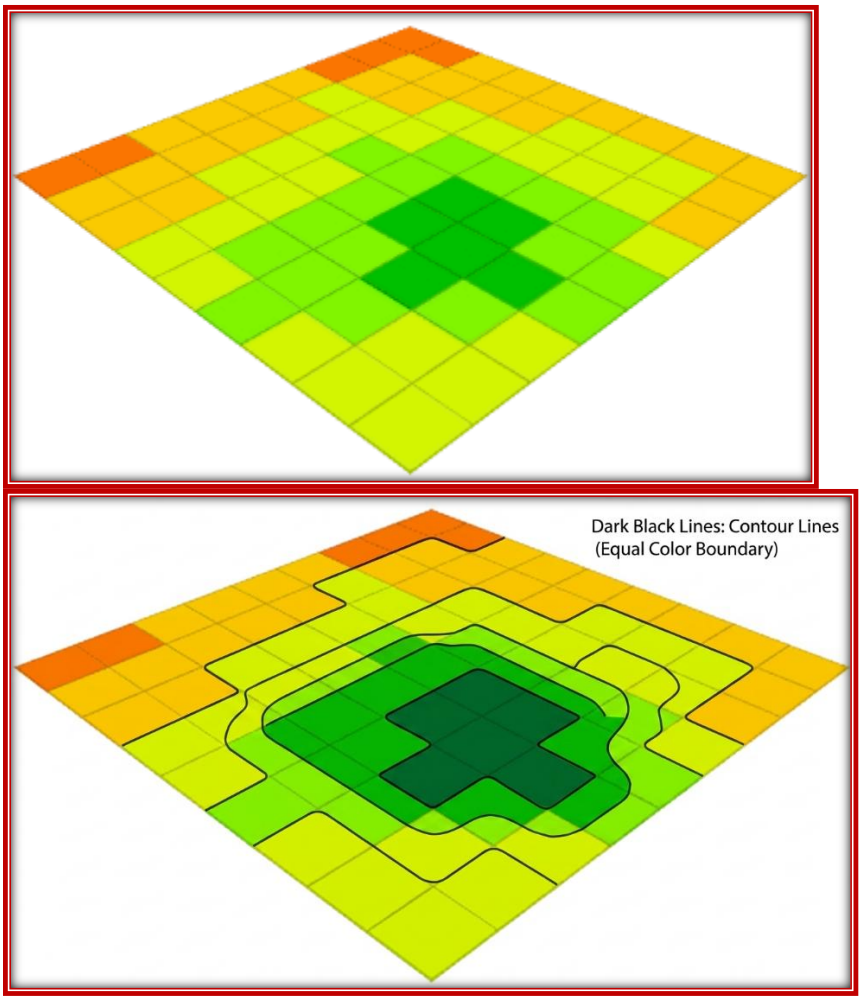
Lab Settings

Use Data in folder: **AGT1-Data/Module1**



Generate Contours

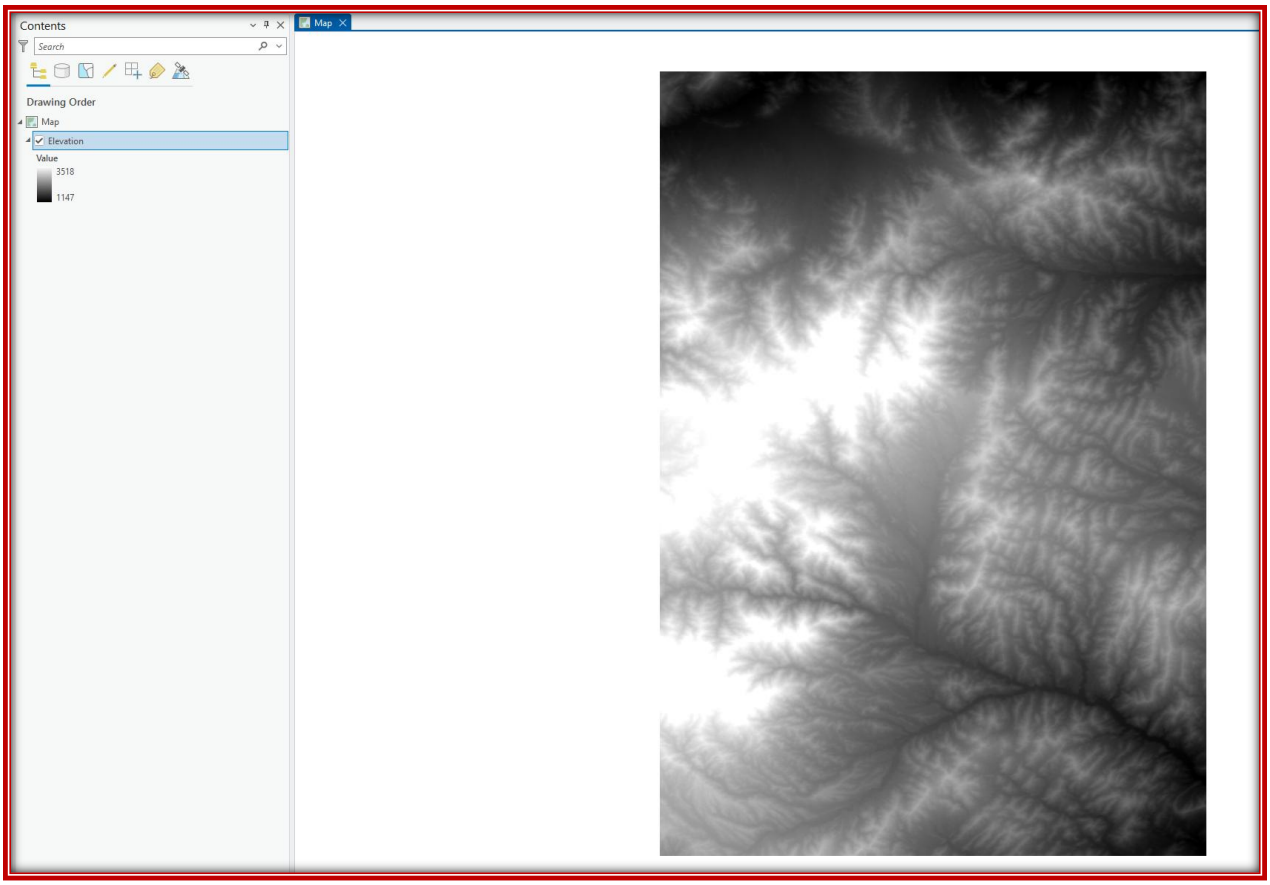
A **contour** is a line on a map that connects points of equal value, such as elevation, temperature, or pressure. It helps visualize how a surface changes over an area.

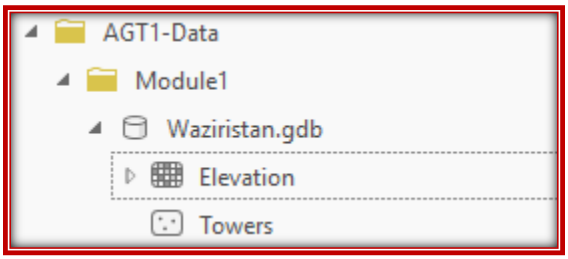


Key Concepts

- **Equal Value:** Every point along a single contour line has the exact same measurement.
- **The Interval:** This is the fixed difference in value between two adjacent contour lines (e.g., every 10 feet or 5 degrees).
  - **Closer lines** indicate a rapid change (a steep slope).
  - **Wider lines** indicate a gradual change (a flat area).
- **The Base Contour:** This is your starting point (often zero). Contours are drawn at set intervals both above and below this base value.

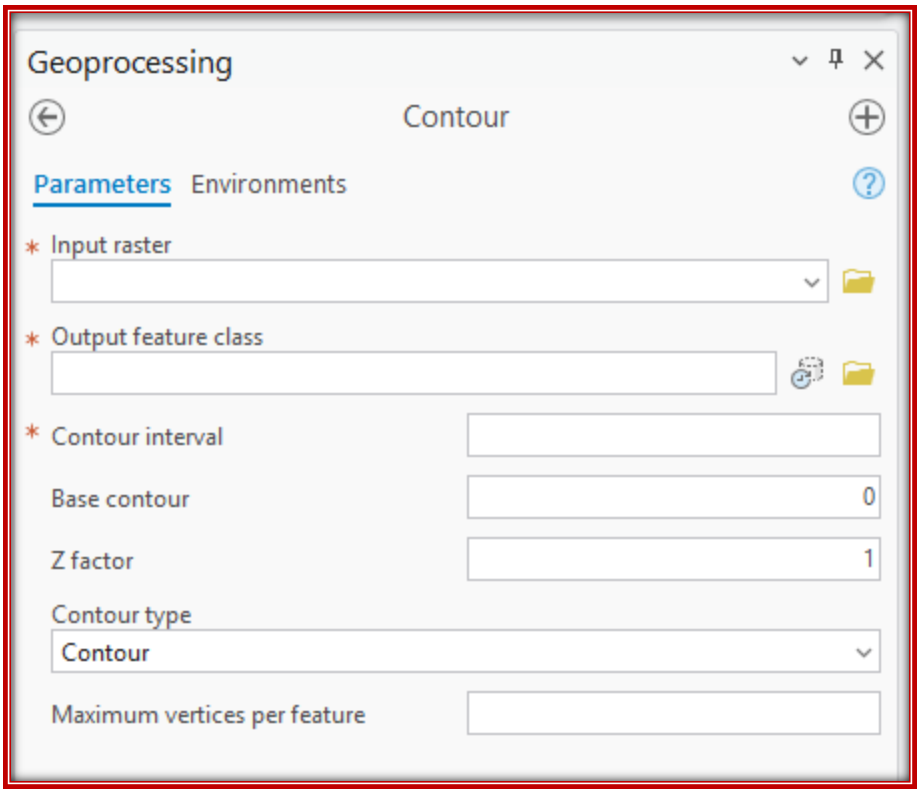
Goal: Generate contours from DEM and highlight index contours





Load Elevation raster (DEM) from Waziristan.gdb

Open Contour Tool **(Spatial Analyst)** and configure parameters



The fields marked with a red star (*) are mandatory.

1. Input raster* This is your starting data. Think of it as a grid (like a spreadsheet of squares) where each square contains a number, such as elevation in feet.

- **What it does:** The tool reads this data to understand the shape of the surface (the hills and valleys).
- **Impact:** A more detailed (higher resolution) input will allow you to create more precise contours.

2. Output feature class* this is simply the file location and name for your new contour map.

- **What it does:** It tells the computer where to save the lines it creates.
- **Impact:** This determines where you will find your map afterward. Be sure to use a clear name so you remember what it is!

3. Contour interval* This is the single most important parameter. It determines the "steepness" or vertical jump from one line to the next.

- **Low Value (e.g., 2 meters):** This will create a **high-detail** map. It will have many lines drawn very close together. This is great for showing fine details but can look extremely messy and "noisy" in rugged or flat terrain.
- **High Value (e.g., 50 meters):** This will create a **low-detail** map. It will have very few lines, providing a simple, clean overview of the area. However, it will completely hide small hills and intricate details.

4. Base contour This is your "starting line" or reference point. The tool will draw lines at set intervals both above and below this specific value.

- **Default Value (0):** This is the most common setting, especially for maps based on sea level. It results in clean, predictable line values like 100, 200, 300.
- **A Non-Zero Value (e.g., 77):** Your lines will no longer be round numbers. If your interval is 10 and your base is 77, your lines would be at ..., 67, 77, 87, 97, ... This is incredibly useful if you need a contour to lie precisely on a known boundary value (like a flood stage elevation or property boundary) and don't care about "nice" round numbers for the other lines.

5. Z factor This is a simple multiplier used for converting units or correcting horizontal vs. vertical measurements. The real terrain doesn't change; only the numbers drawn on the map change.

- **1 (Default):** No conversion. The output values are the same as the input values.
- **Multiplier Value:** The output values will be multiplied by this number.
    - *Example (Meters to Feet):* If your data is in **meters** and you want lines labeled in **feet**, you would use a Z factor of **3.2808**. A hill that is 1000m high in the data would have a contour line labeled "3280.8".
    - *Example (Feet to Meters):* For the reverse, you would use a Z factor of **0.3048**.

6. Contour type This determines the type of geometric shape the tool will create.

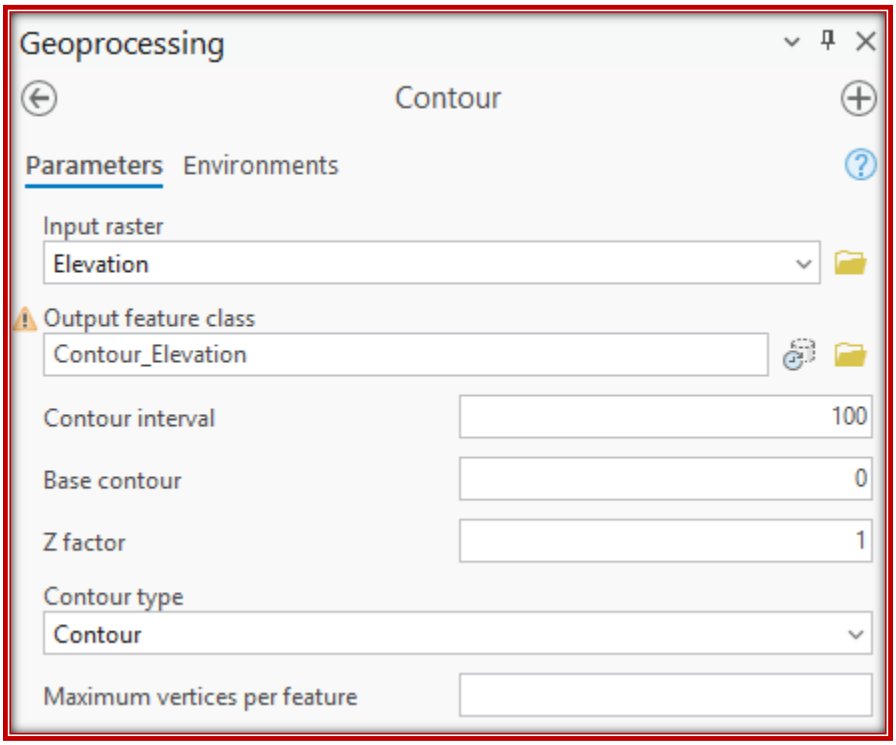
- **Contour (Default):** This creates **lines**. Every point along that line has the same elevation. This is the standard view for a topographic map.
- **Contour Polygon:** This creates **enclosed shapes** (like puddles). Every point *inside* a particular polygon has a value that is *between* the values of the lines that define it. This is more commonly used for analytical purposes rather than traditional mapping.

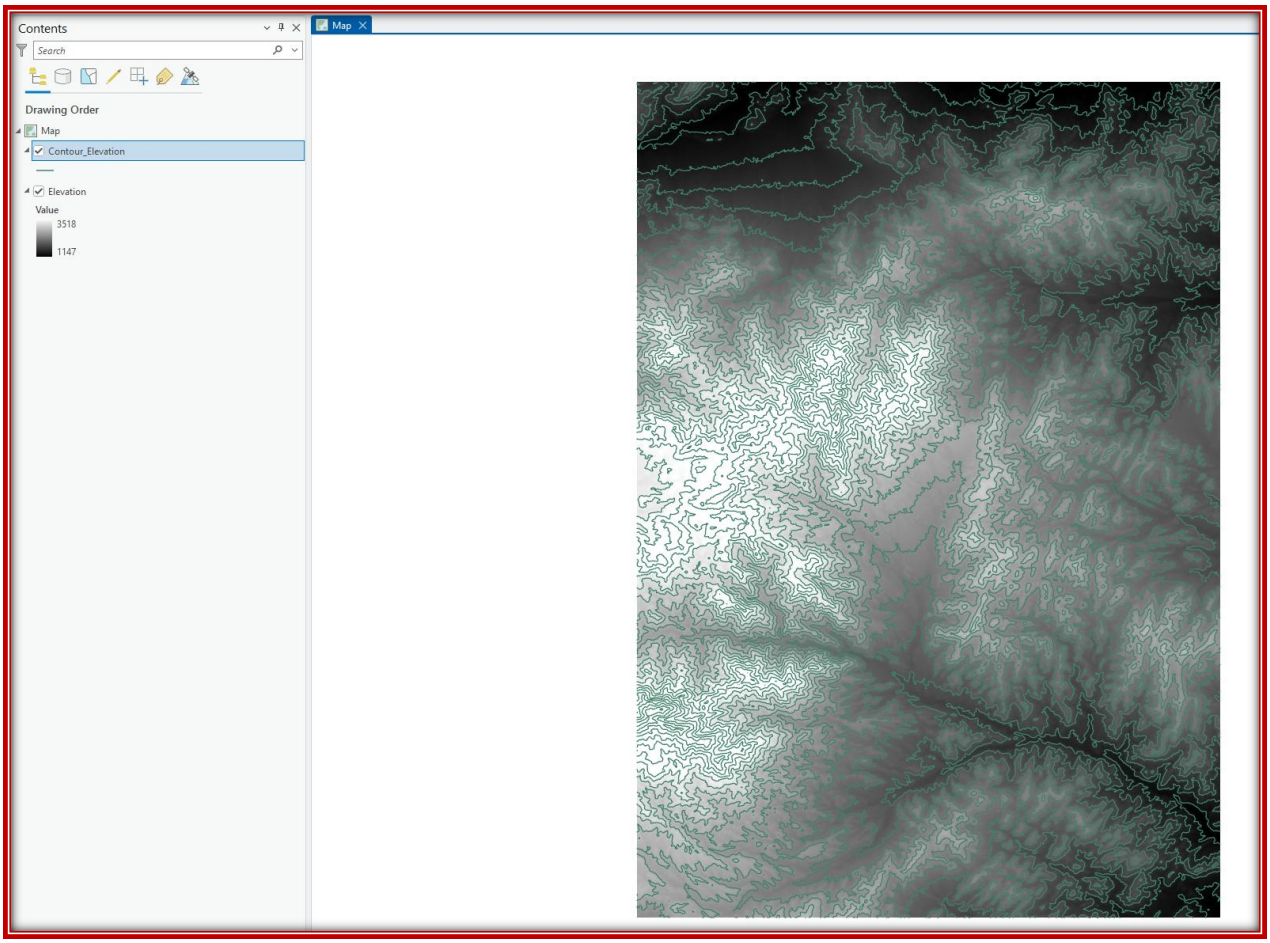
7. Maximum vertices per feature This is a technical limit on how complex a single contour line can be.

- **Blank (Default):** No limit. The computer will draw one continuous line for a single elevation value, even if it has thousands of twists and points.
- **Small Number (e.g., 100):** If a single line becomes too complex (like in extremely rugged terrain), the computer will "chop" it into multiple smaller

segments. Each segment will still have the same elevation value, but they will be separate "pieces." This makes the final file easier to work with, edit, and analyze in other parts of the GIS.

For this exercise set parameters as follows and hit run:





Contours have been generated.

Next step is to highlight every 5th contour (the index contour).

We shall follow two method to achieve this.

Method 1:

To do this, we check attributes of generated contours:

Contour_Elevation

Field: Add Calculate Selection: Select By At

	OBJECTID *	Shape *	Id	Contour	Shape_Length
1	1	Polyline	1	1200	0.002862
2	2	Polyline	2	1200	0.001244
3	3	Polyline	3	1200	0.000378
4	4	Polyline	4	1200	0.001215
5	5	Polyline	5	1200	0.007086
6	6	Polyline	6	1200	0.070292
7	7	Polyline	7	1200	0.029135
8	8	Polyline	8	1300	0.001003
9	9	Polyline	9	1300	0.013142
10	10	Polyline	10	1300	0.001781
11	11	Polyline	11	1300	0.022977
12	12	Polyline	12	1300	0.000188
13	13	Polyline	13	1300	0.002335
14	14	Polyline	14	1300	0.00077
15	15	Polyline	15	1300	0.003013
16	16	Polyline	16	1300	0.000809
17	17	Polyline	17	1300	0.000876
18	18	Polyline	18	1300	0.020059
19	19	Polyline	19	1300	0.303872
20	20	Polyline	20	1300	0.000301

We add a new field called **“IsIndex”** to identify index contours

Contour_Elevation | *Fields: Contour_Elevation

Current Layer: Contour_Elevation

Visible	Read Only	Field Name	Alias	Data Type	Allow NULL	Highlight	Number Format	Domain	Default	Length
✓	✓	OBJECTID	OBJECTID	Object ID			Numeric			
✓		Shape	Shape	Geometry						
✓		Id	Id	Long			Numeric			
✓		Contour	Contour	Double			Numeric			
✓	✓	Shape_Length	Shape_Length	Double	✓		Numeric			
✓		IsIndex		Long	✓					

Click here to add a new field.

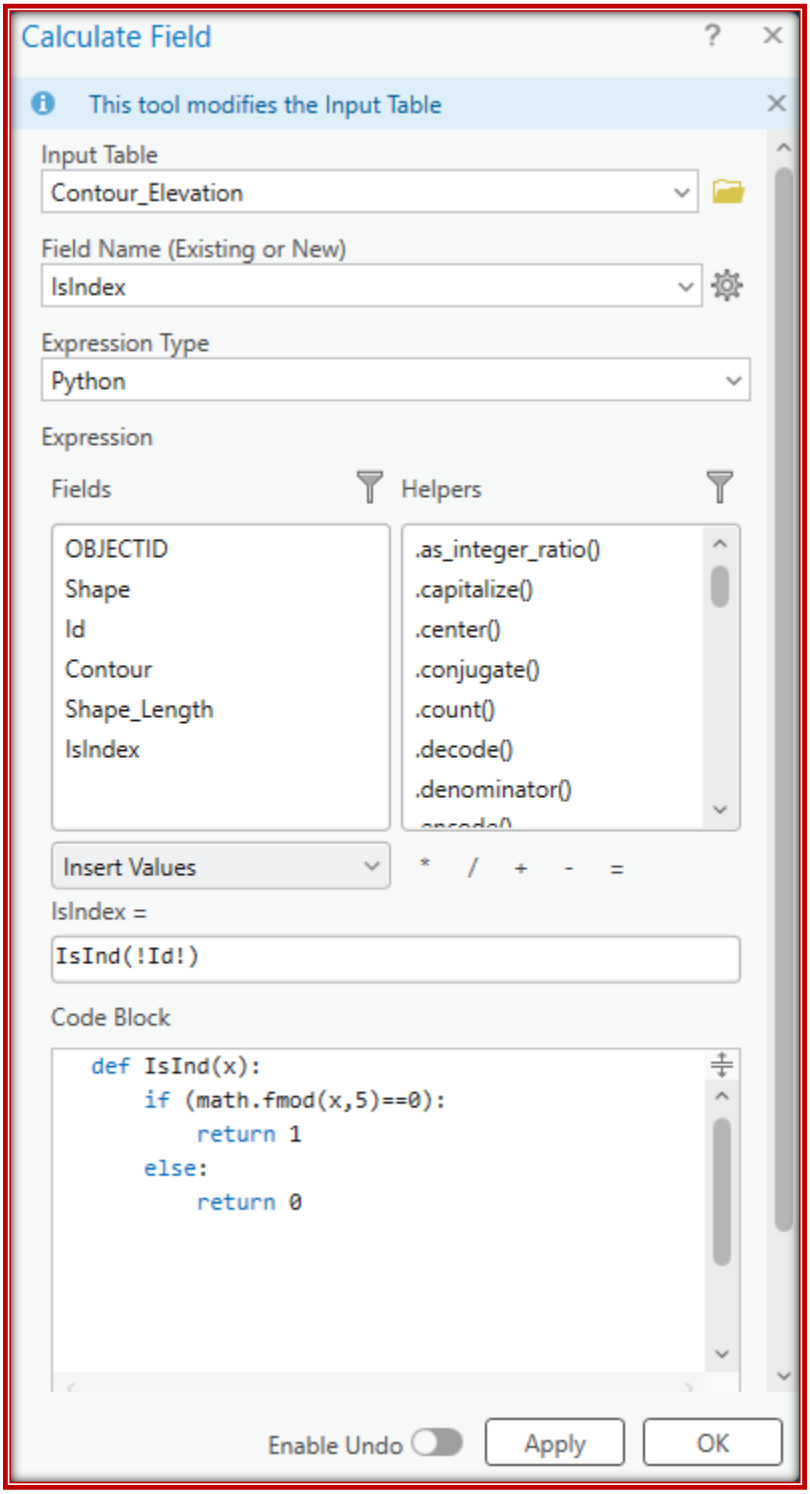
Contour_Elevation

Field: Add Calculate | Selection: Select By Attributes Zoom

	OBJECTID *	Shape *	Id	Contour	Shape_Length	IsIndex
1	1	Polyline	1	1200	0.002862	<Null>
2	2	Polyline	2	1200	0.001244	<Null>
3	3	Polyline	3	1200	0.000378	<Null>
4	4	Polyline	4	1200	0.001215	<Null>
5	5	Polyline	5	1200	0.007086	<Null>
6	6	Polyline	6	1200	0.070292	<Null>
7	7	Polyline	7	1200	0.029135	<Null>
8	8	Polyline	8	1300	0.001003	<Null>
9	9	Polyline	9	1300	0.013142	<Null>
10	10	Polyline	10	1300	0.001781	<Null>
11	11	Polyline	11	1300	0.022977	<Null>
12	12	Polyline	12	1300	0.000188	<Null>
13	13	Polyline	13	1300	0.002335	<Null>
14	14	Polyline	14	1300	0.00077	<Null>
15	15	Polyline	15	1300	0.003013	<Null>
16	16	Polyline	16	1300	0.000809	<Null>
17	17	Polyline	17	1300	0.000876	<Null>
18	18	Polyline	18	1300	0.020059	<Null>
19	19	Polyline	19	1300	0.303872	<Null>
20	20	Polyline	20	1300	0.000301	<Null>

0 of 1,058 selected

We can then use a **python function** and **math.modf()** to set **IsIndex** to 1 for every 5th contour:



Think of this code in this code as a tiny "sorting machine" that tags special lines for your map.

The Code Block:

- **def**: Short for "define." It’s like saying, "I am building a new machine."
- **Function (IsInd)**: This is the name of the machine. It is a set of stored instructions that performs a specific task so you don't have to do the math yourself.
- **x**: This is a placeholder. It represents the "raw material" (a number) you drop into the machine.

Variables:

- **!Id!**: The exclamation marks tell the computer to go look at your table and grab the value from the actual **Id** column for the current row.
- **IsIndex**: This is the "label" the machine creates.

The Step-by-Step Logic

1. **Input**: The computer grabs a number from the **!Id!** column.
2. **The Test (math.fmod())**: It checks if that number is divisible by **5** with no remainder.
3. **The Output**:

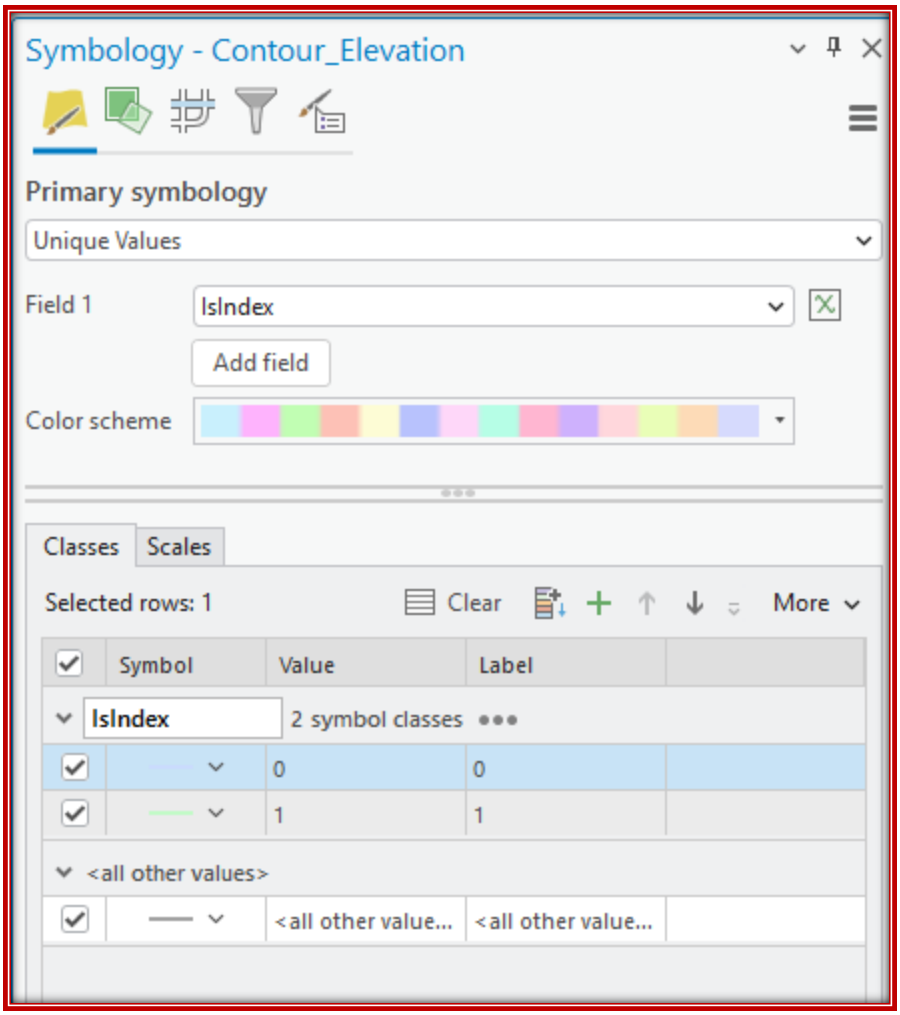
    - If it fits perfectly, it gives you a **1** (Yes).
    - If there is a remainder, it gives you a **0** (No).
    - You now have a column of 1s and 0s that tells the map exactly which lines to make thicker (the index contours).

Contour_Elevation

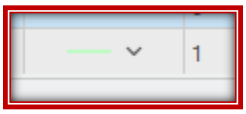
Field: Add Calculate Selection: Select By Attribute

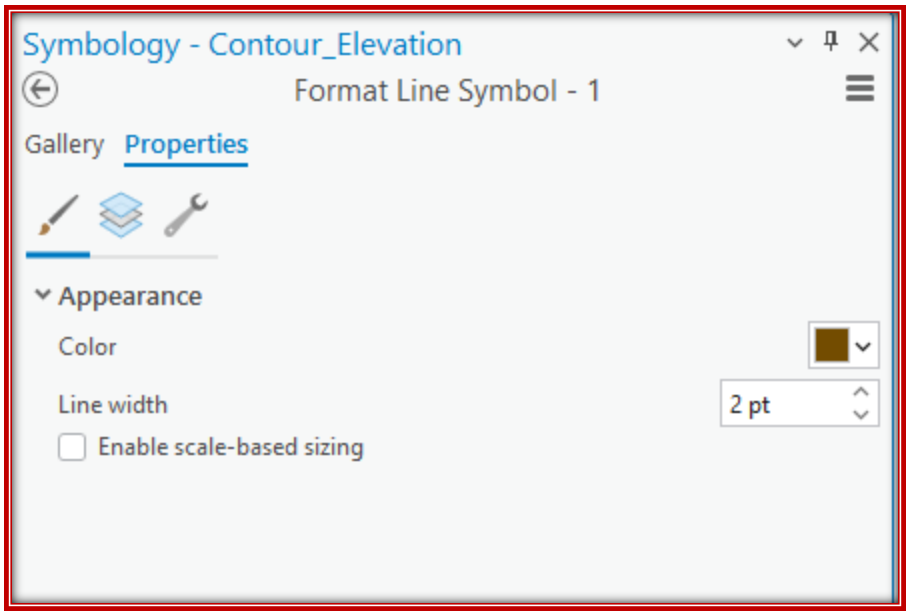
	OBJECTID *	Shape *	Id	Contour	Shape_Length	IsIndex
1	1	Polyline	1	1200	0.002862	0
2	2	Polyline	2	1200	0.001244	0
3	3	Polyline	3	1200	0.000378	0
4	4	Polyline	4	1200	0.001215	0
5	5	Polyline	5	1200	0.007086	1
6	6	Polyline	6	1200	0.070292	0
7	7	Polyline	7	1200	0.029135	0
8	8	Polyline	8	1300	0.001003	0
9	9	Polyline	9	1300	0.013142	0
10	10	Polyline	10	1300	0.001781	1
11	11	Polyline	11	1300	0.022977	0
12	12	Polyline	12	1300	0.000188	0
13	13	Polyline	13	1300	0.002335	0
14	14	Polyline	14	1300	0.00077	0
15	15	Polyline	15	1300	0.003013	1
16	16	Polyline	16	1300	0.000809	0
17	17	Polyline	17	1300	0.000876	0
18	18	Polyline	18	1300	0.020059	0
19	19	Polyline	19	1300	0.303872	0
20	20	Polyline	20	1300	0.000301	1

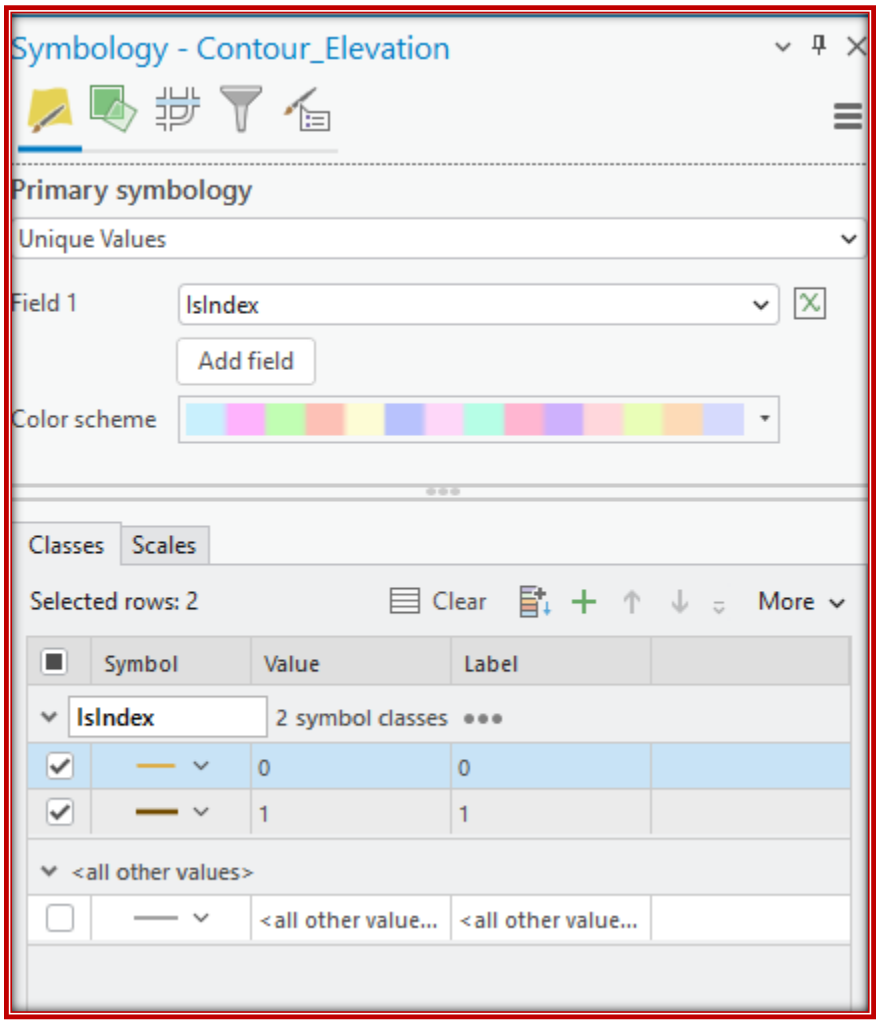
Now that Index contours have **IsIndex** set to 1, we can apply unique symbology based on this field:



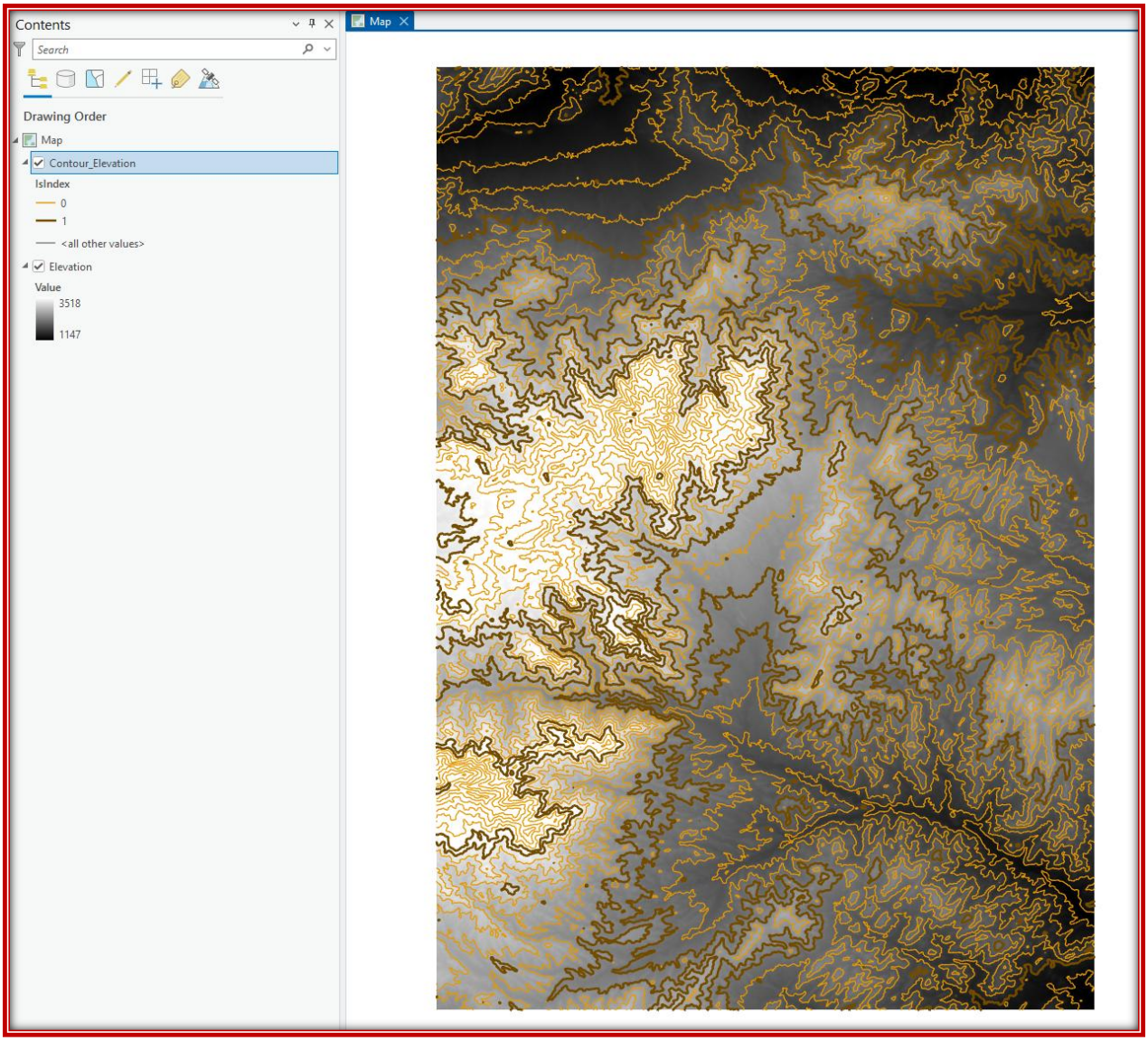
Clicking on each symbol, we can go to properties and change color/width:




Symbology - Contour_Elevation
Format Line Symbol - 1
Gallery
Properties
Appearance
Color
Line width
2 pt
Enable scale-based sizing


Symbology - Contour_Elevation
Primary symbology
Unique Values
Field 1
IsIndex
Add field
Color scheme
Classes
Scales
Selected rows: 2
Clear
More
Symbol
Value
Label
IsIndex
2 symbol classes
0
0
1
1
<all other values>
<all other value...
<all other value...

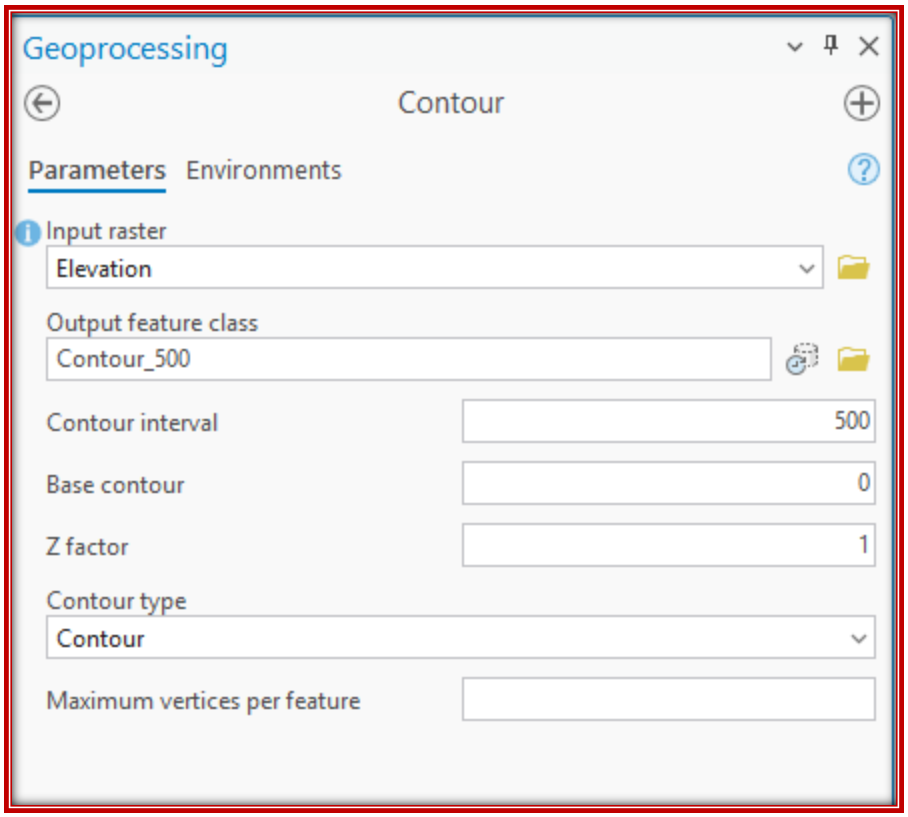
This should return contours like:

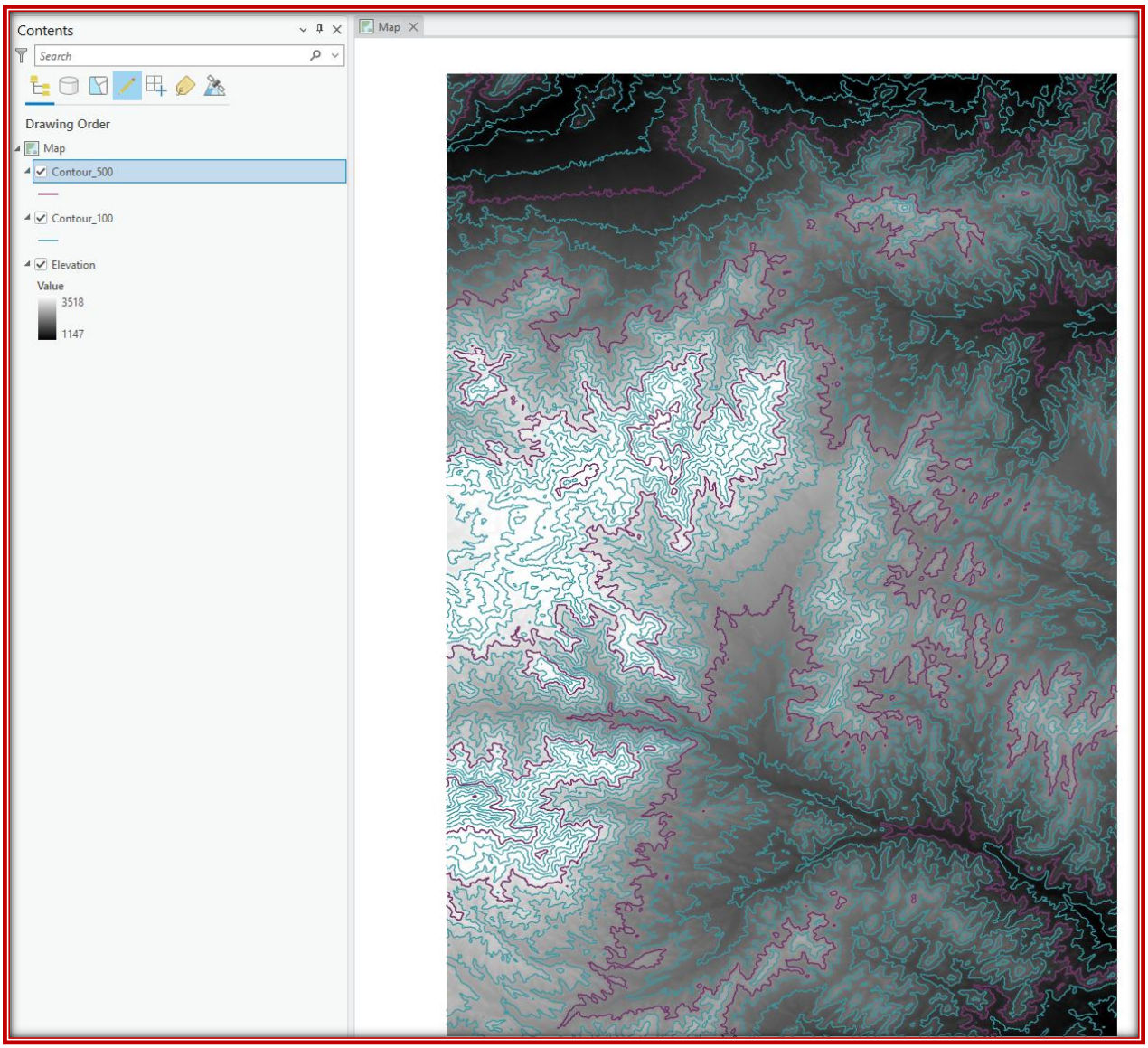


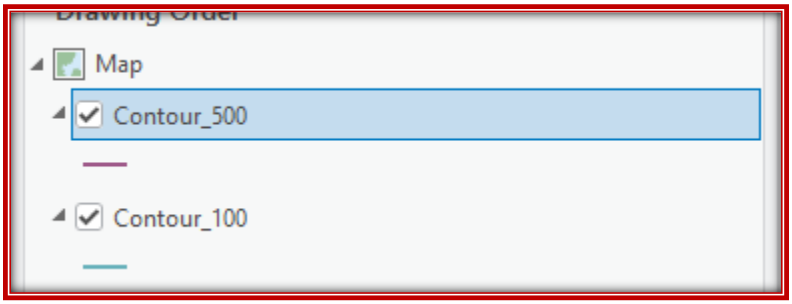
Method 2:

An alternative is to generate new contours with interval set to every **5th contour**.

For example, in our example we set contour interval to **100m**. This means that every 5th contour is **500m** apart. We simply make an new set of contours with **contour interval of 500:**


Geoprocessing
Contour
Parameters
Environments
Input raster
Elevation
Output feature class
Contour_500
Contour interval
500
Base contour
0
Z factor
1
Contour type
Contour
Maximum vertices per feature





This is a **crude** approach, not recommended for professional GIS analysts such as yourself, which generates two layers (redundant), but it is good enough for presentation purpose.

Generate Hillshade

Hillshading is a mapping technique that simulates how sunlight hits a surface to make flat maps look three-dimensional. It calculates brightness values (from 0 to 255) for every spot based on the sun's position.

The Goal is to create a realistic background that makes terrain "pop".

Mapmakers usually place the "sun" in the **North-West** (upper-left).

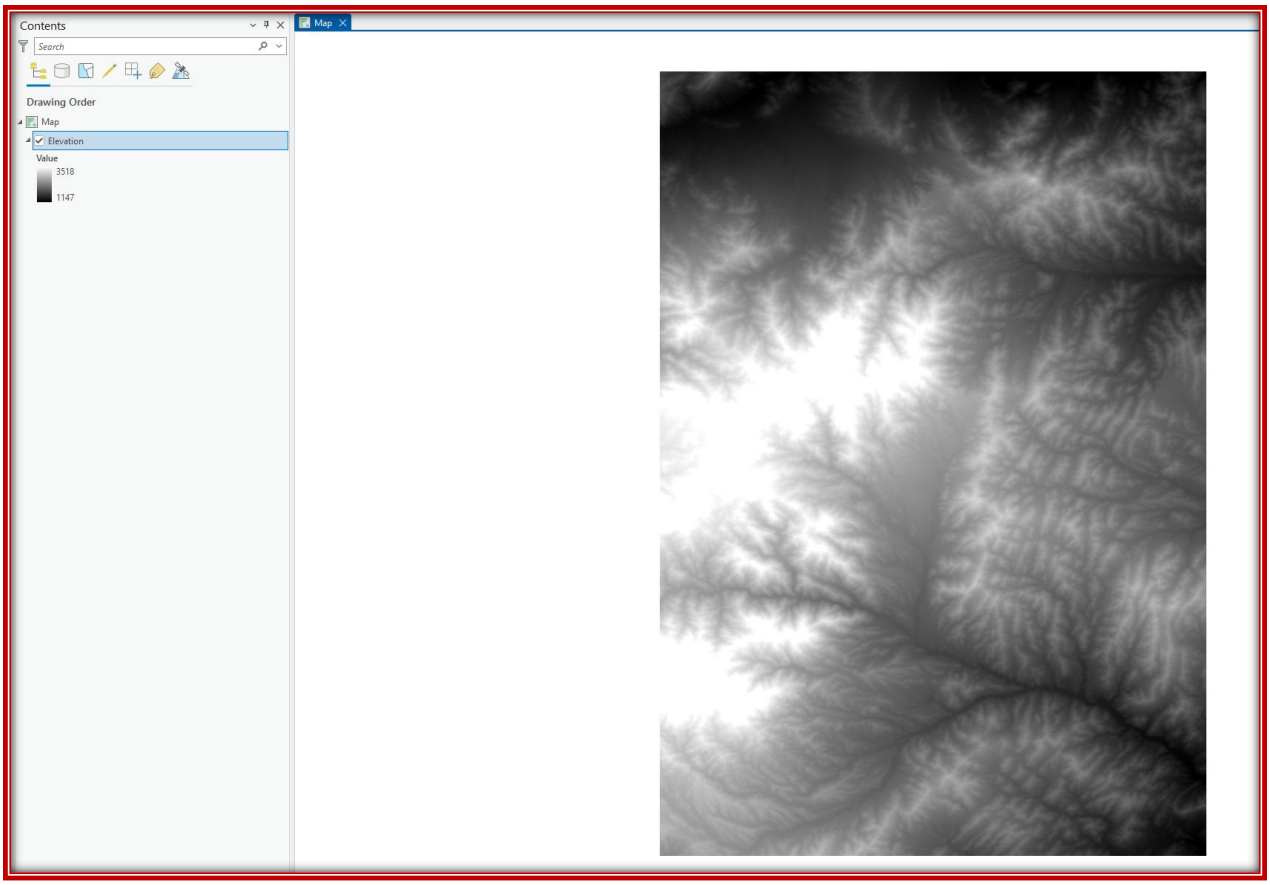
Why North-West?

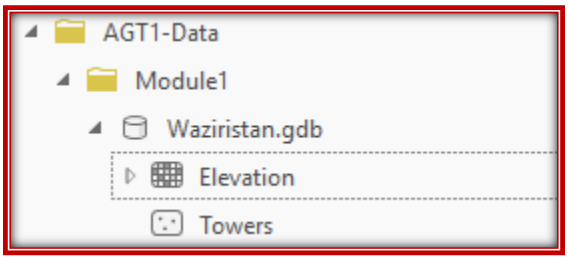
Our eyes expect shadows at the bottom of objects; if the light comes from elsewhere, mountains can look like holes in the ground.

Scientific Use: To measure actual sun exposure for things like solar panel placement. You must use the "real" sun position (azimuth and altitude) for a specific time and geographic location.

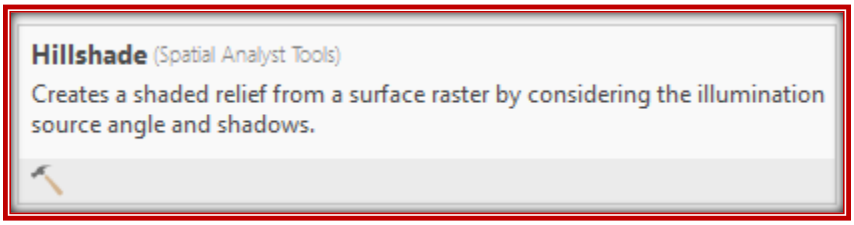
Higher values (closer to 255) indicate areas getting the most direct light.

Goal: Generate hillshade of our DEM to better visualize it.

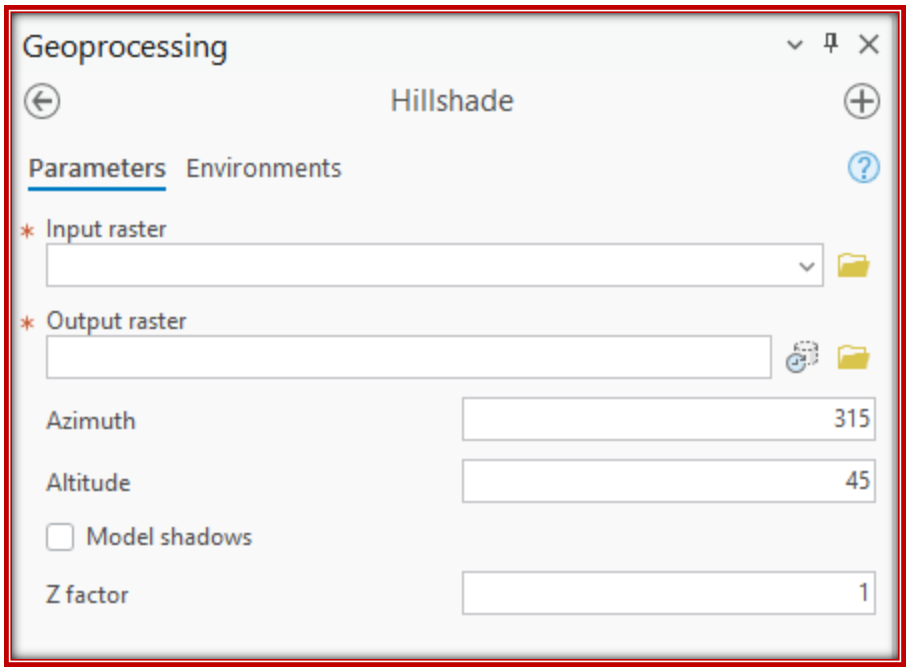




Load Elevation raster (DEM) from Waziristan.gdb



Open **Hillshade** (Spatial Analyst Tool) and configure parameters:



- **Input raster***: This is your elevation data (like a DEM) that provides the "shape" of the ground.
- **Output raster***: The name and location for your new hillshade image file.

Lighting Parameters

These values determine where your "fake sun" is sitting in the sky.

1. Azimuth (Compass Direction)

This is the compass direction the sun is shining from, measured in degrees from 0 to 360.

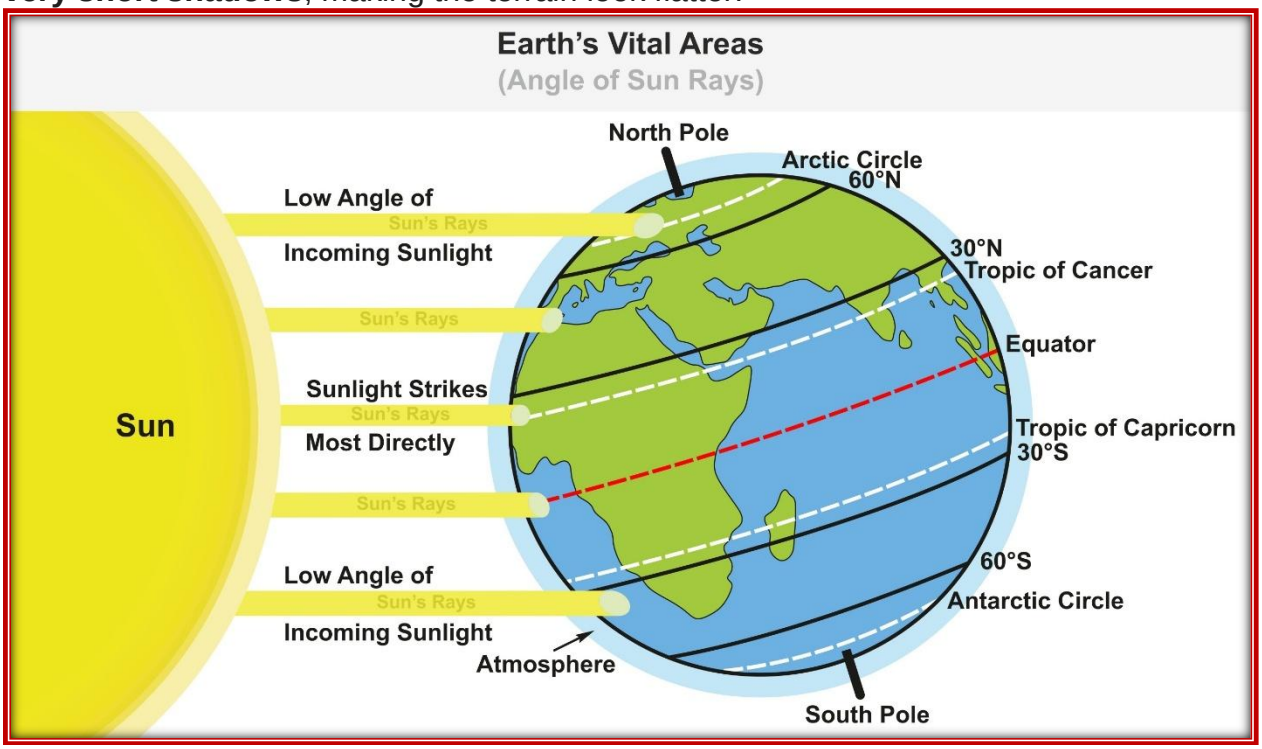
- **315 (Default):** This places the light in the **North-West**. It is the standard for maps because it prevents mountains from looking like holes.
- **90:** The light comes from the **East**. This would make the eastern slopes bright and cast long shadows to the west.
- **180:** The light comes from the **South**.

2. Altitude (Height in the Sky)

This is the angle of the sun above the horizon, from 0 to 90 degrees.

- **45 (Default):** A balanced height that creates clear, natural-looking shadows.

- **Low Value (e.g., 10):** The sun is near the horizon (like sunset). This creates **very long, dramatic shadows** and high contrast.
- **High Value (e.g., 80):** The sun is almost directly overhead (like noon). This creates **very short shadows**, making the terrain look flatter.



Shutterstock

Advanced Parameters

3. Model shadows

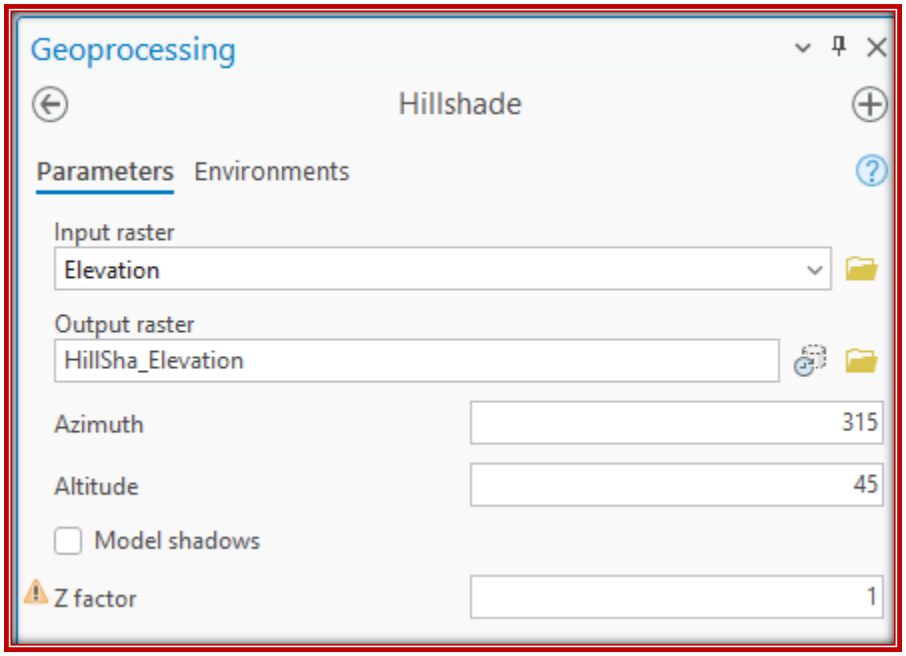
- **Unchecked (Default):** The tool only calculates "local" shading based on which way a slope faces.
- **Checked:** The tool also calculates "cast shadows." If a giant mountain is tall enough, it will actually cast a dark shadow over the valley behind it. This is more realistic but can make the map look very dark.

4. Z factor

This is a multiplier used to make your height units match your map units.

- **1 (Default):** No change to the data.
- **Adjustment:** If your map is in **meters** but your elevation data is in **feet**, you use this to fix the math so the hills don't look 3x taller or shorter than they actually are.

For this exercise we use the following parameters:



IMPORTANT BEFORE YOU HIT RUN!!!

We need to configure environment for this tool as well to make sure we use appropriate coordinate system. This is important because setting your map to a **GCS** or **PCS** gives different results because they use different units for math.

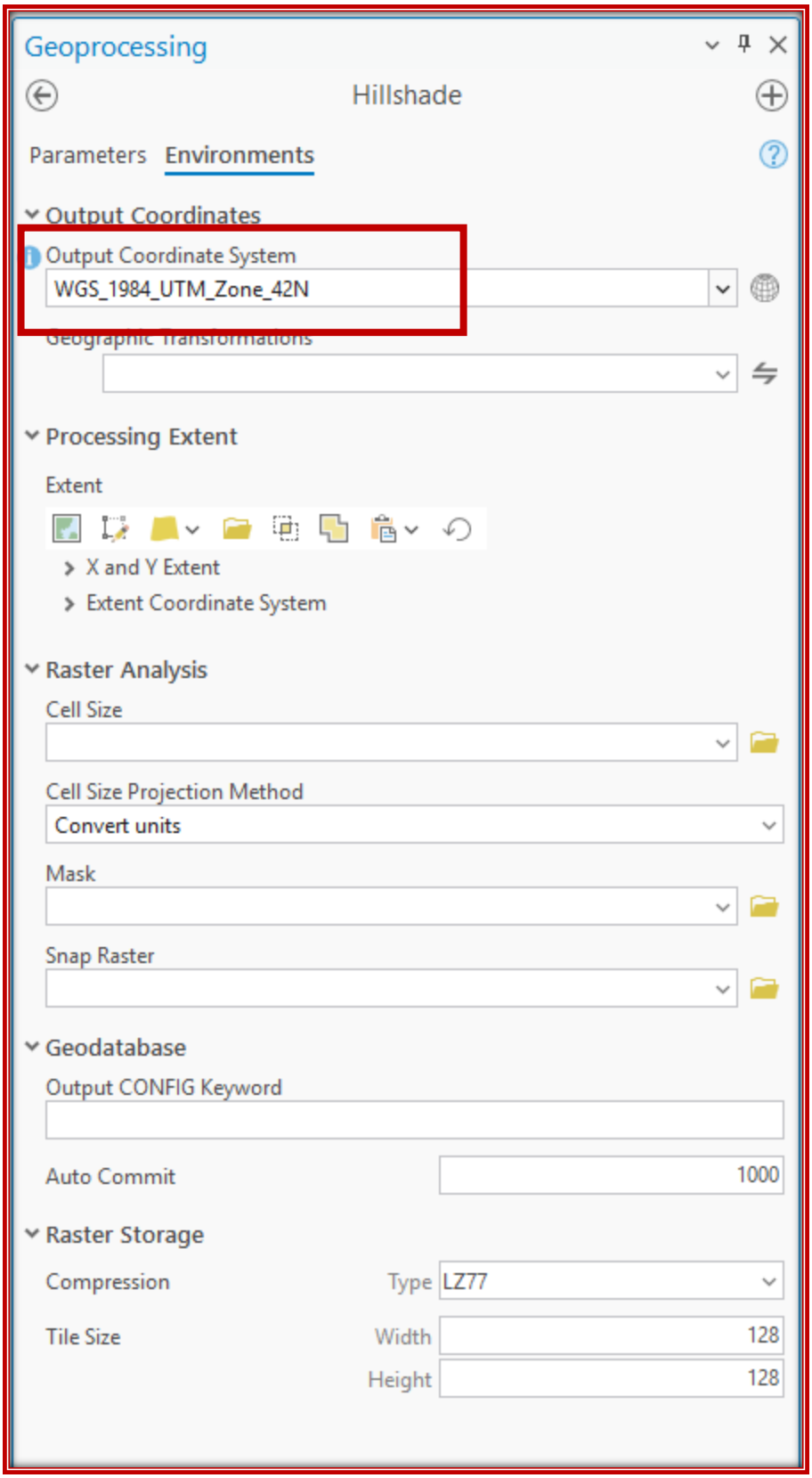
GCS (Geographic) = Degrees

- **The Unit:** Measures the Earth in **degrees**.
- **The Problem:** Degrees are huge compared to meters.
- **The Result:** The computer gets confused trying to compare "0.0001 degrees" (sideways) to "500 meters" (up). Your map will look **flat or gray** because the math is out of sync.

PCS (Projected) = Meters

- **The Unit:** Measures the Earth in **meters or feet**.
- **The Benefit:** Both horizontal and vertical distances use the **same unit**.
- **The Result:** The math is "honest". Shadows look deep and realistic because a 1-meter rise is compared to a 1-meter run.

Summary: Always use a **PCS** (like UTM) for Hillshades and Contours to keep your units consistent and your map looking 3D.


Geoprocessing
Hillshade
Parameters
Environments
Output Coordinates
Output Coordinate System
WGS_1984_UTM_Zone_42N
Geographic Transformations
Processing Extent
Extent
X and Y Extent
Extent Coordinate System
Raster Analysis
Cell Size
Cell Size Projection Method
Convert units
Mask
Snap Raster
Geodatabase
Output CONFIG Keyword
Auto Commit
1000
Raster Storage
Compression
Type
LZ77
Tile Size
Width
128
Height
128

1. Output Coordinate System

This defines the "map rules" (the grid) the new file will live on.

- **What it does**: It ensures your hillshade lines up perfectly with your other map layers (like roads or rivers).
- **Different Values**: If you choose a system like **UTM Zone 42N** (shown in the image), the map will be optimized for a specific part of the world. If you leave it blank, it usually just copies the system from your input file.

2. Processing Extent

Think of this as a **cookie cutter**.

- **What it does**: It limits the area the tool actually processes.
- **Different Values**:
  - **Default**: It processes everything in your input file.
  - **Specific Area**: You can set it to only process the area you can currently see on your screen, which saves time if your original file is massive.

3. Raster Analysis

This controls the "resolution" and shape of your output.

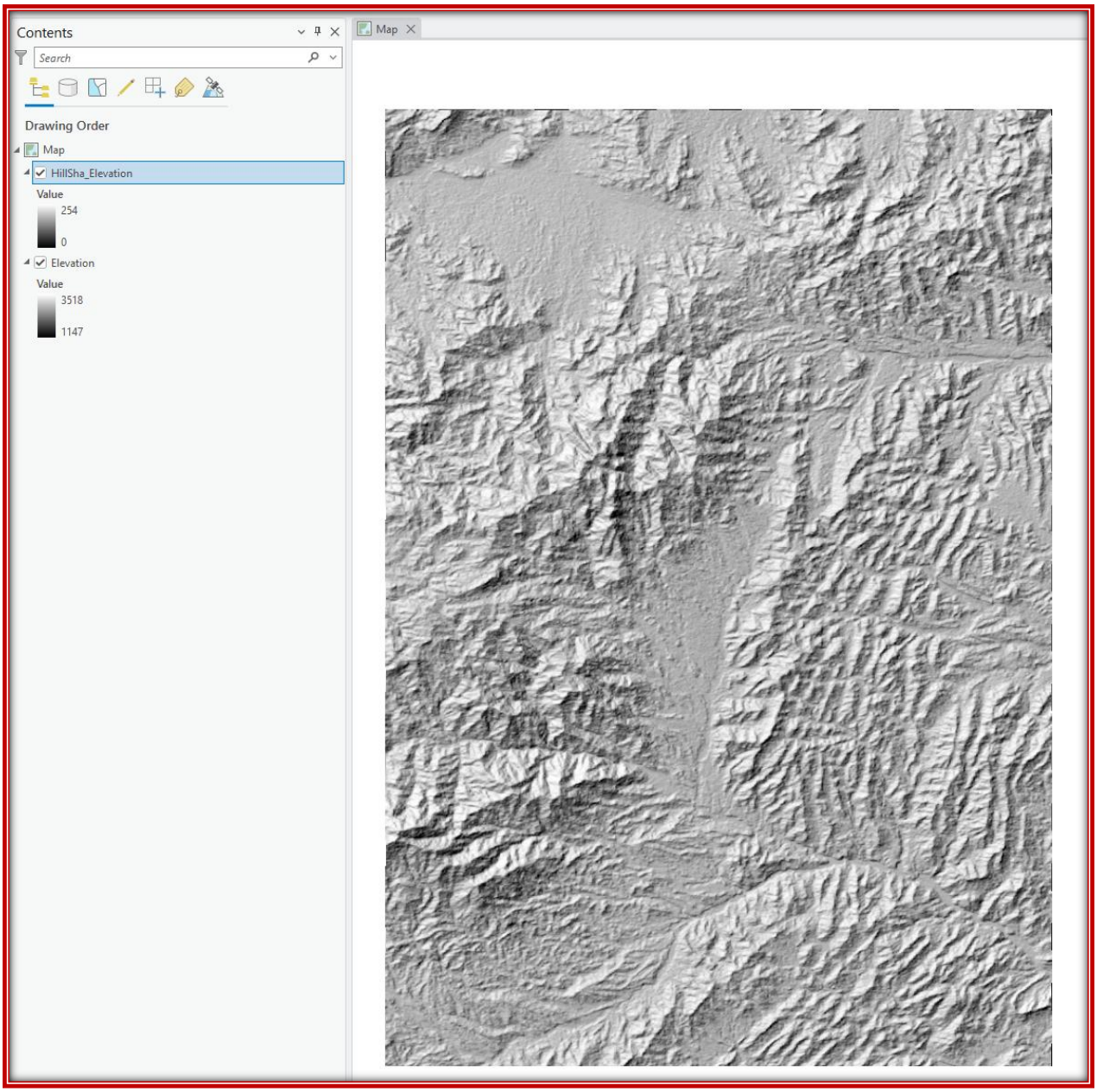
- **Cell Size**: This is the size of the tiny squares (pixels) that make up your map.
  - **Small Value**: Makes a very sharp, detailed map (but creates a huge file).
  - **Large Value**: Makes a "blocky" or blurry map (but runs very fast).
- **Mask**: This is like **painter's tape**.
  - If you select a shape (like a county boundary), the tool will only draw the hillshade *inside* that shape and leave everything else empty.

4. Raster Storage (Compression)

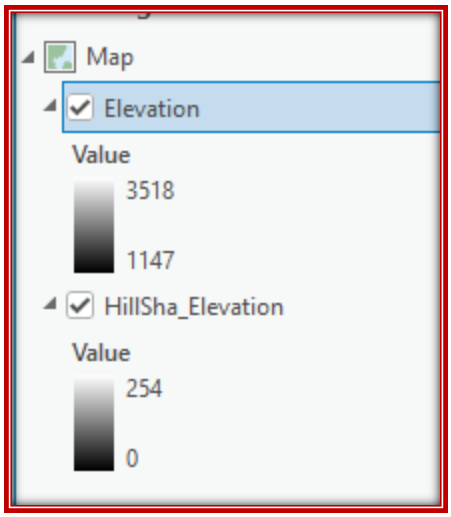
This manages the **file size**.

- **Compression Type (LZ77)**: This is like a ZIP file for your image.
  - **What it does**: It squishes the data down so it takes up less space on your hard drive without losing any quality.
- **Tile Size (128x128)**: This breaks the image into small "chunks."
  - This helps the computer load the map faster by only opening the chunks you are currently looking at.

This should give result like:

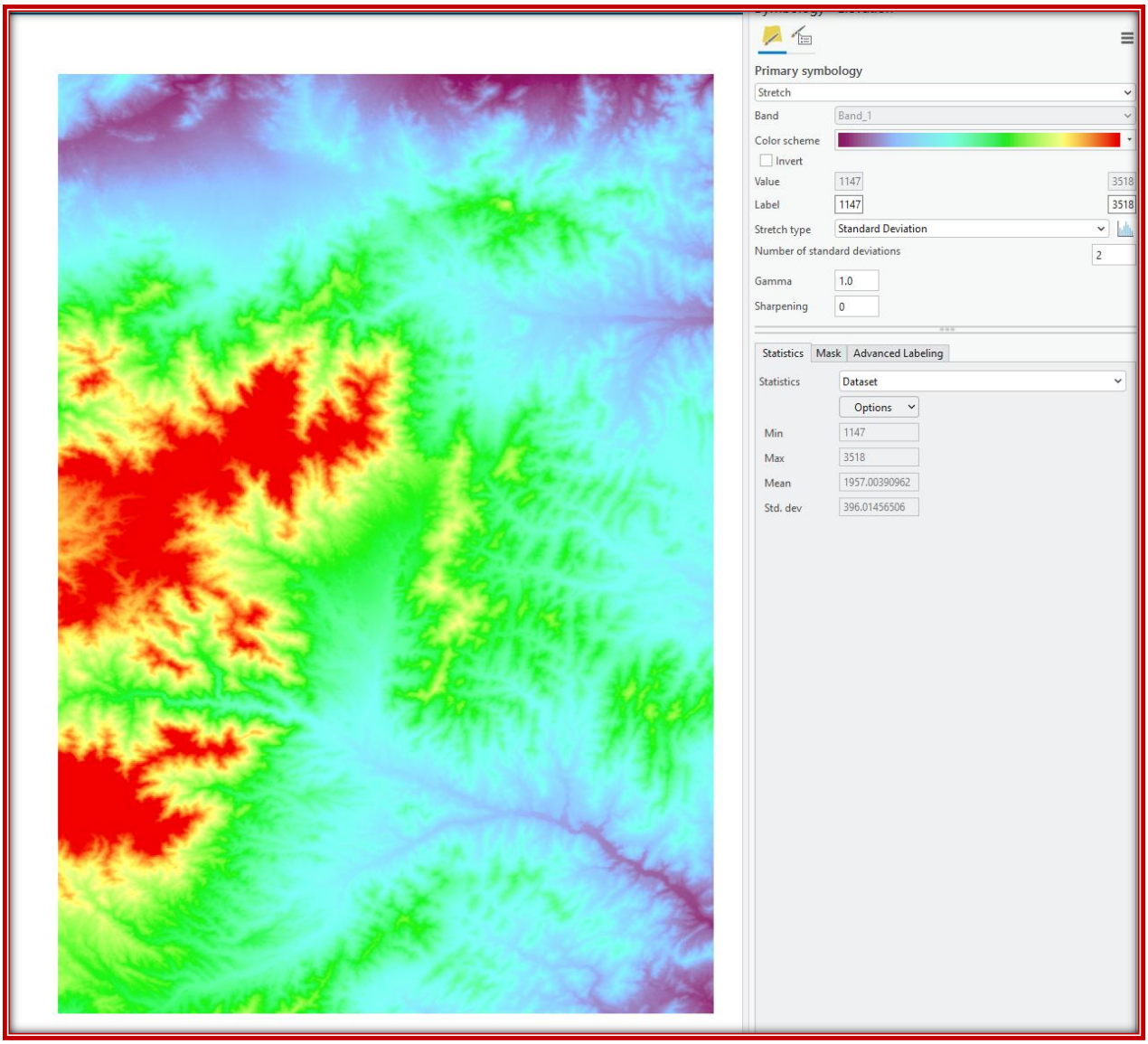


Although good but black & white does not look good enough. In order to improve the **visualization** we can overlay the DEM with appropriate symbology.

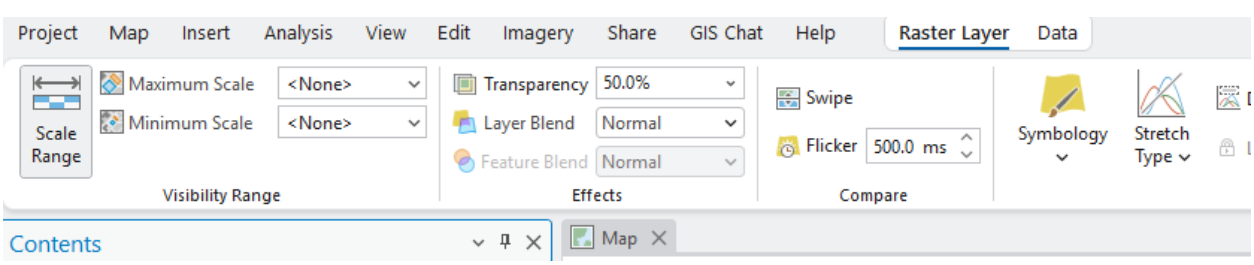


Drag **Elevation layer (DEM raster)** on top of the hillshade.

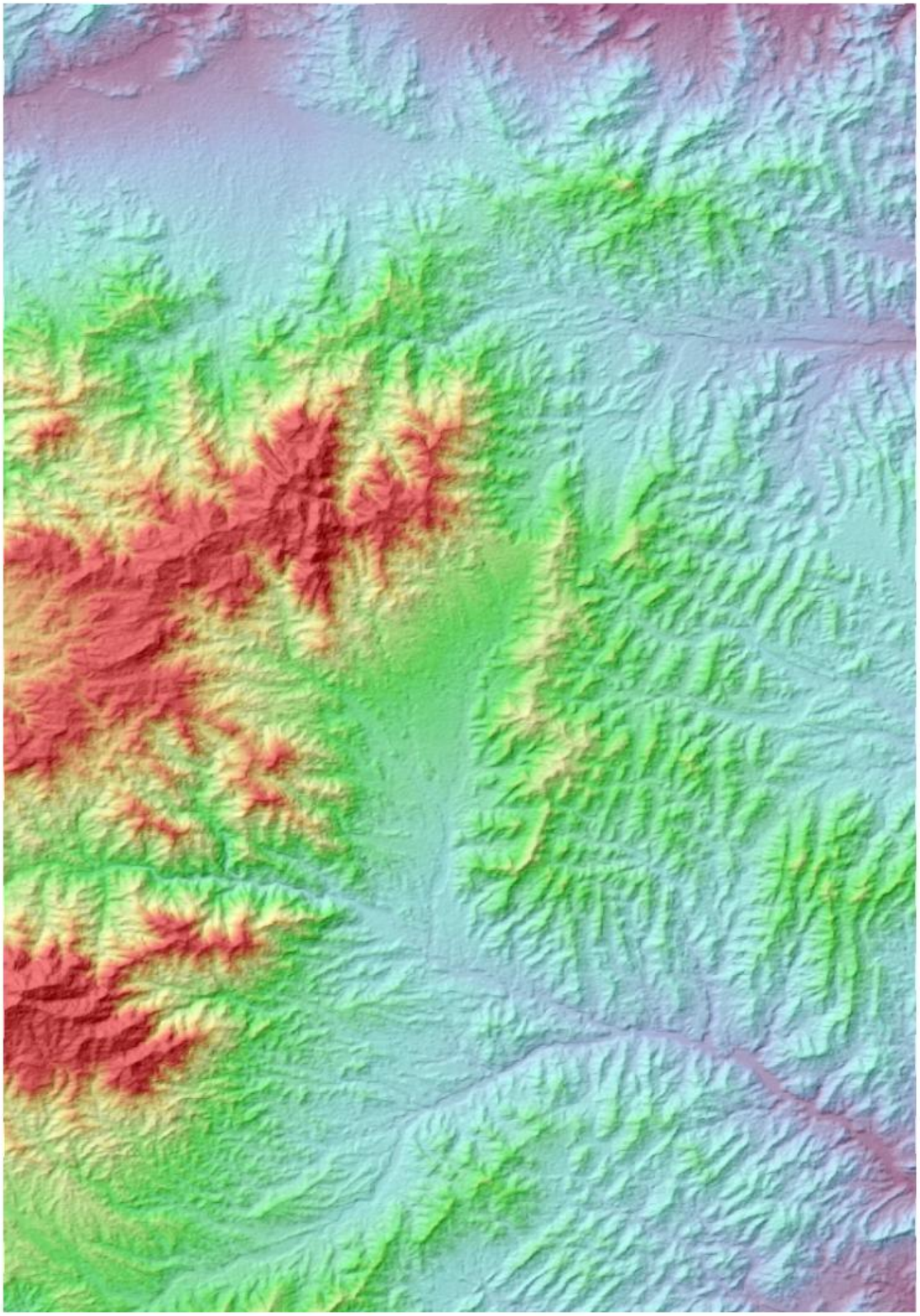
Change symbology of this layer to an appropriate color:



Go to raster layer tab for the **DEM** and set transparency to 50%:



This should return a better looking map:

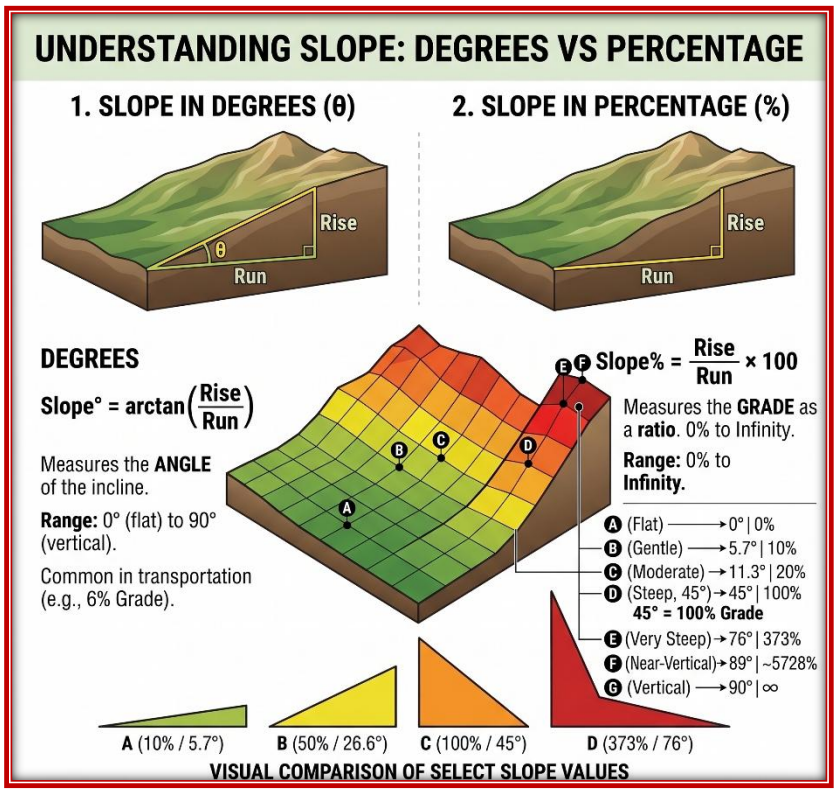


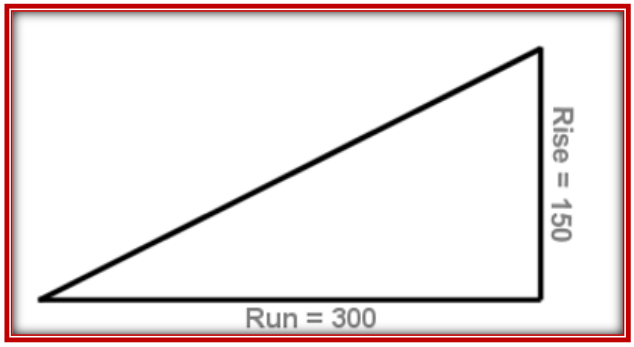
Calculate Slope

In GIS, slope represents the steepness or incline of a surface, typically measured as the maximum rate of change in value between a cell and its neighbors.

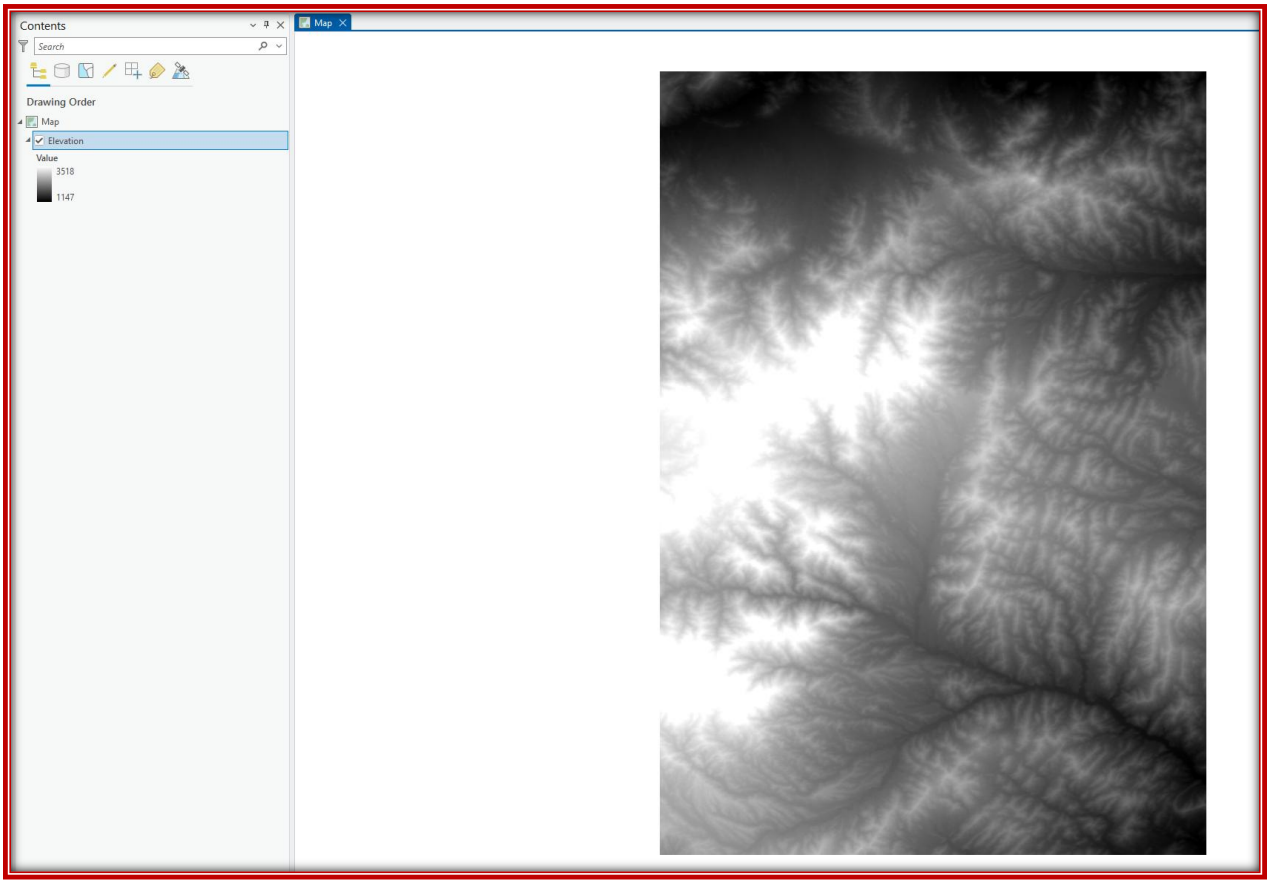
It is a key measure of a surface's morphology, revealing how quickly elevation (the z-value) changes over a specific distance. You can express slope in **degrees** (0 to 90)—which is common in scientific applications, or as a **percentage**, often used in transportation to describe road grades.

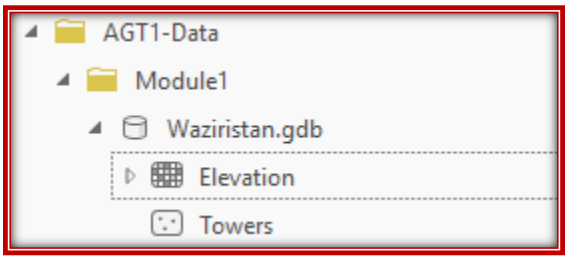
While most often applied to topography for erosion or construction analysis, slope can also be used to track the intensity of change in other data layers, such as rainfall.





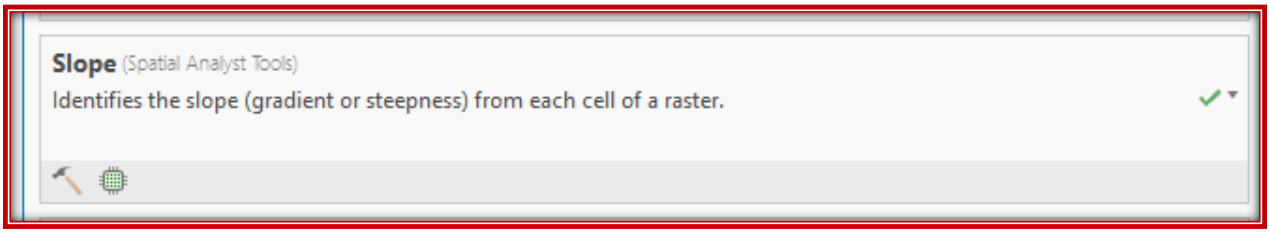
Goal: Calculate slope of our DEM.

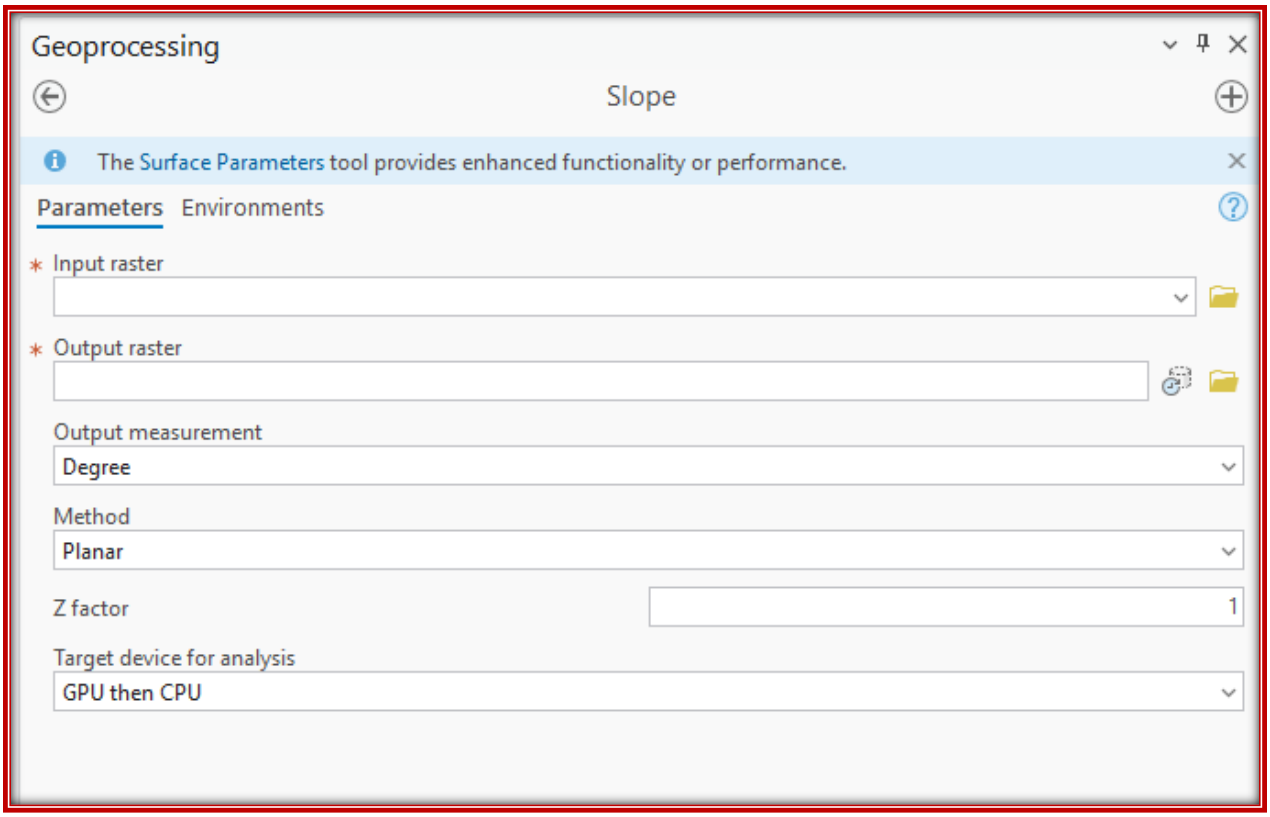




Load Elevation raster (DEM) from Waziristan.gdb

Open **Slope** tool (spatial analyst):





Input raster: The source elevation surface (typically a Digital Elevation Model or DEM).

Output raster: The name and location for the new raster file that will contain the calculated slope values.

Output measurement: Defines the units for the output slope:

> **Degree:** Calculates the angle of inclination (0 to 90 degrees).
>
> **Percent rise:** Calculates the "rise over run" multiplied by 100 (0 to essentially infinity).

Method: Determines the geometry used for the calculation:

> **Planar:** Performs the calculation on a flat 2D plane. It is faster but less accurate over large areas or at high latitudes.
>
> **Geodesic:** Accounts for the actual curvature of the Earth (the ellipsoid). This is more accurate for datasets in geographic coordinates.

Z factor: A scaling factor used if your vertical (elevation) units are different from your horizontal (x,y) units.

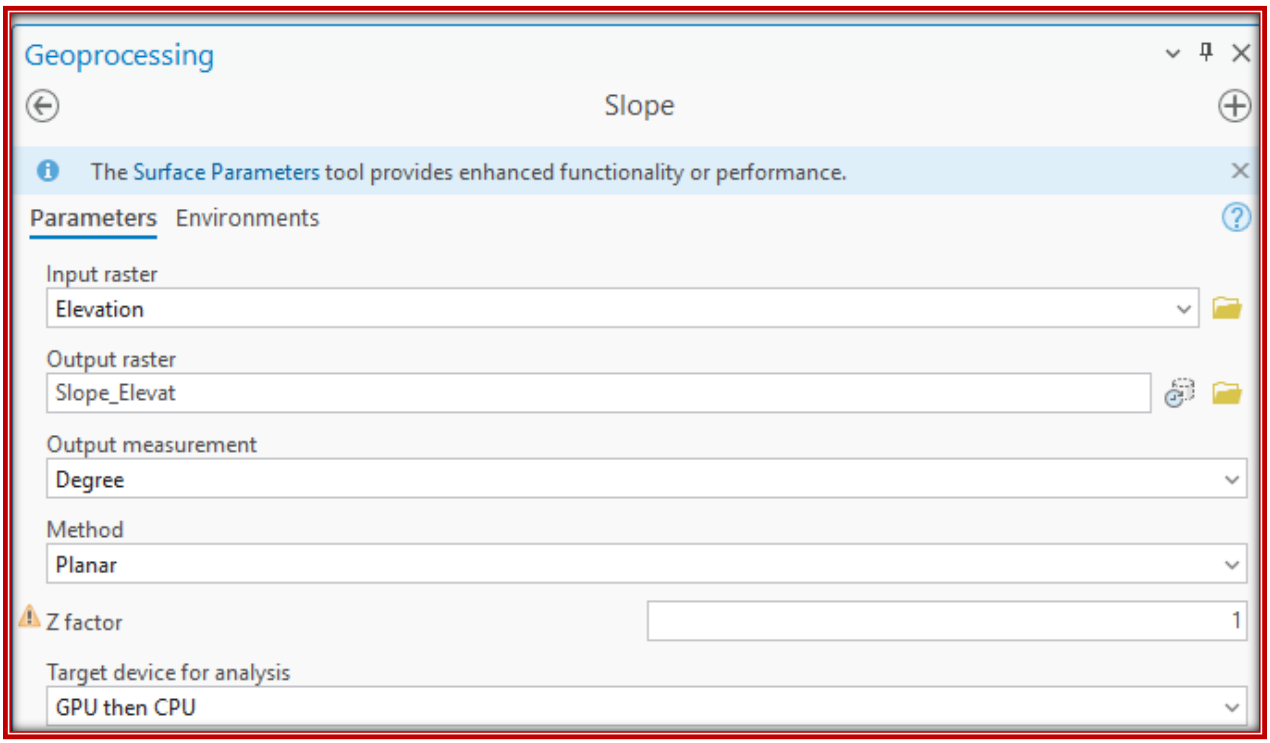
For example, if your map units are meters but your elevation is in feet, you would use a Z factor of 0.3048 to convert feet to meters.

Target device for analysis: Specifies which hardware component will process the tool:

GPU then CPU: Uses the Graphics Processing Unit if available for faster performance; otherwise, it defaults to the CPU.

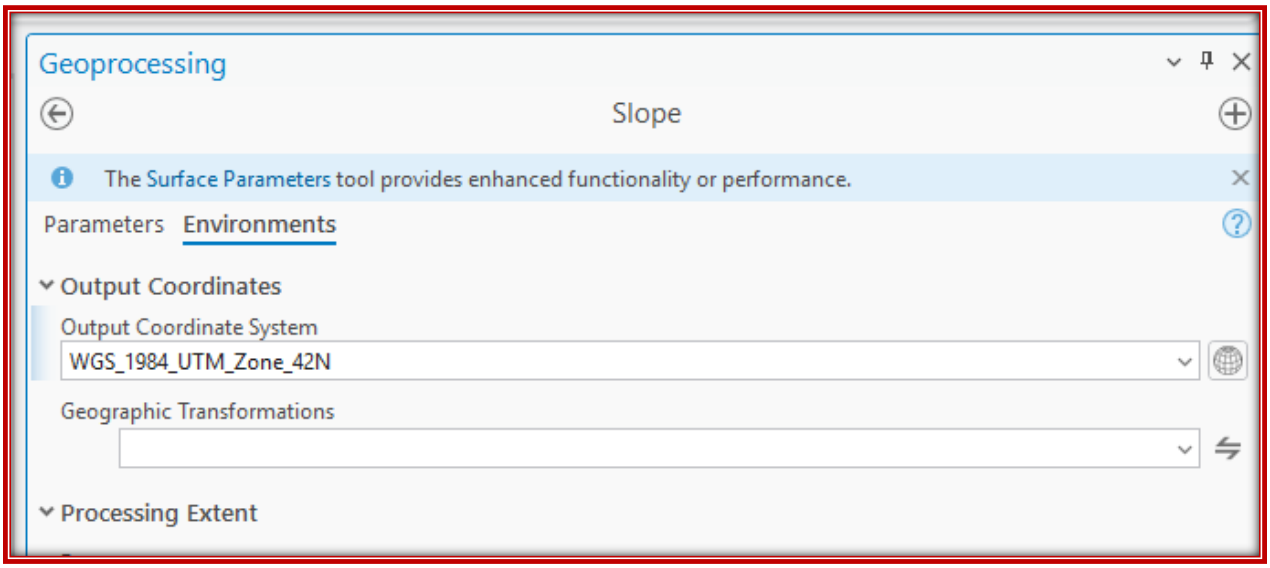
CPU only: Forces the tool to run only on the processor.

Configure it as follows:



IMPORTANT

You must set tool environment to a projected coordinate system



Why?

When calculating slope, using a **Projected Coordinate System (PCS)** like UTM Zone 42N is essential because it ensures that both horizontal (X, Y) and vertical (Z) units are in linear measurements, such as **meters**.

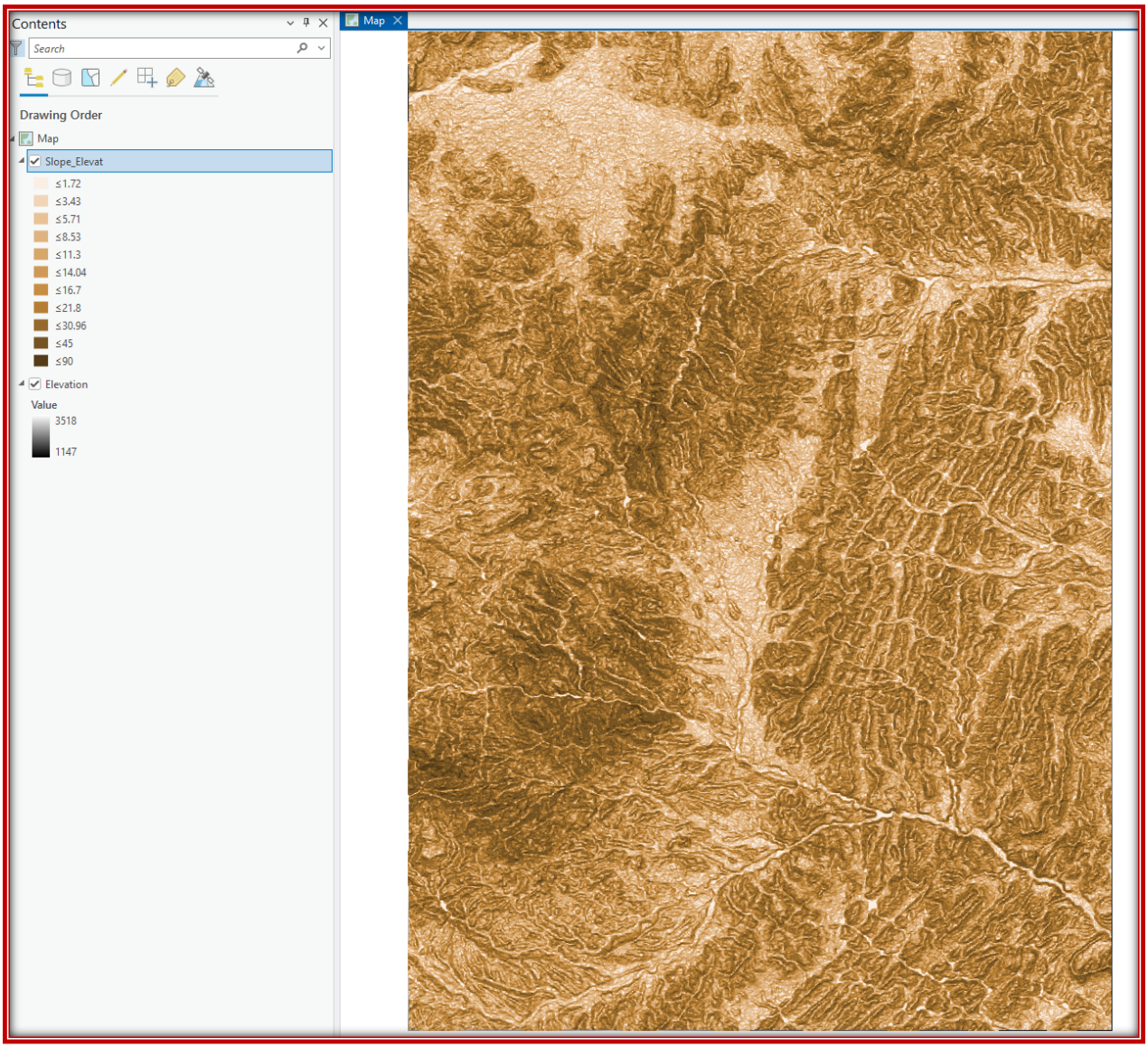
If a **Geographic Coordinate System (GCS)** is used instead, the horizontal units are in **decimal degrees** while the vertical units remain in **meters**.

Because one degree of latitude is roughly **111,320 meters**, the "rise" of the terrain becomes mathematically insignificant compared to the massive "run" of a degree.

This causes the tool to calculate slope values near zero, resulting in a **"very dark"** (try it!) or black raster that incorrectly suggests the terrain is perfectly flat.

Using a PCS aligns these units, allowing the software to accurately represent the steepness of the landscape.

Hit run


Contents
Search
Drawing Order
Map
Slope_Elevat
≤1.72
≤3.43
≤5.71
≤8.53
≤11.3
≤14.04
≤16.7
≤21.8
≤30.96
≤45
≤90
Elevation
Value
3518
1147
Map

Calculate Aspect

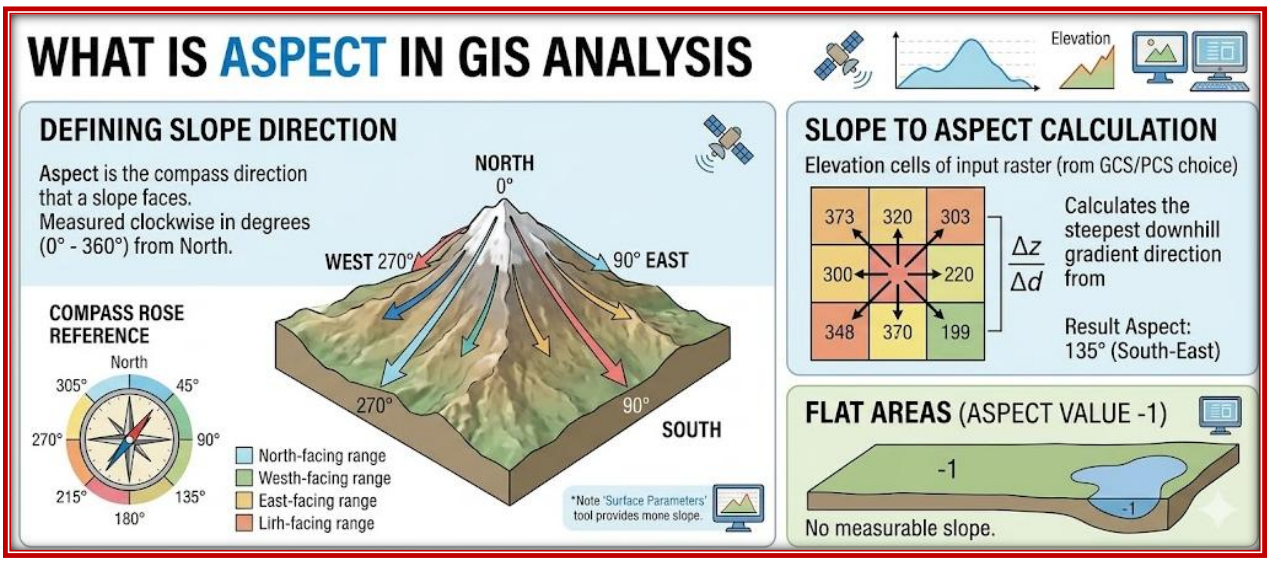
In GIS, Aspect identifies the downslope direction of the maximum rate of change in value from each cell to its neighbors. You can think of it as the compass direction that a slope faces.

- **Measurement:** It is measured clockwise in degrees from 0 (North) to 360 (North again).
  - North: 0° – 22.5° and 337.5° – 360°.
  - East: 67.5° – 112.5°.
  - South: 157.5° – 202.5°.
  - West: 247.5° – 292.5°.
- **Flat Areas:** Any area that has no slope (perfectly level) is assigned a value of -1.

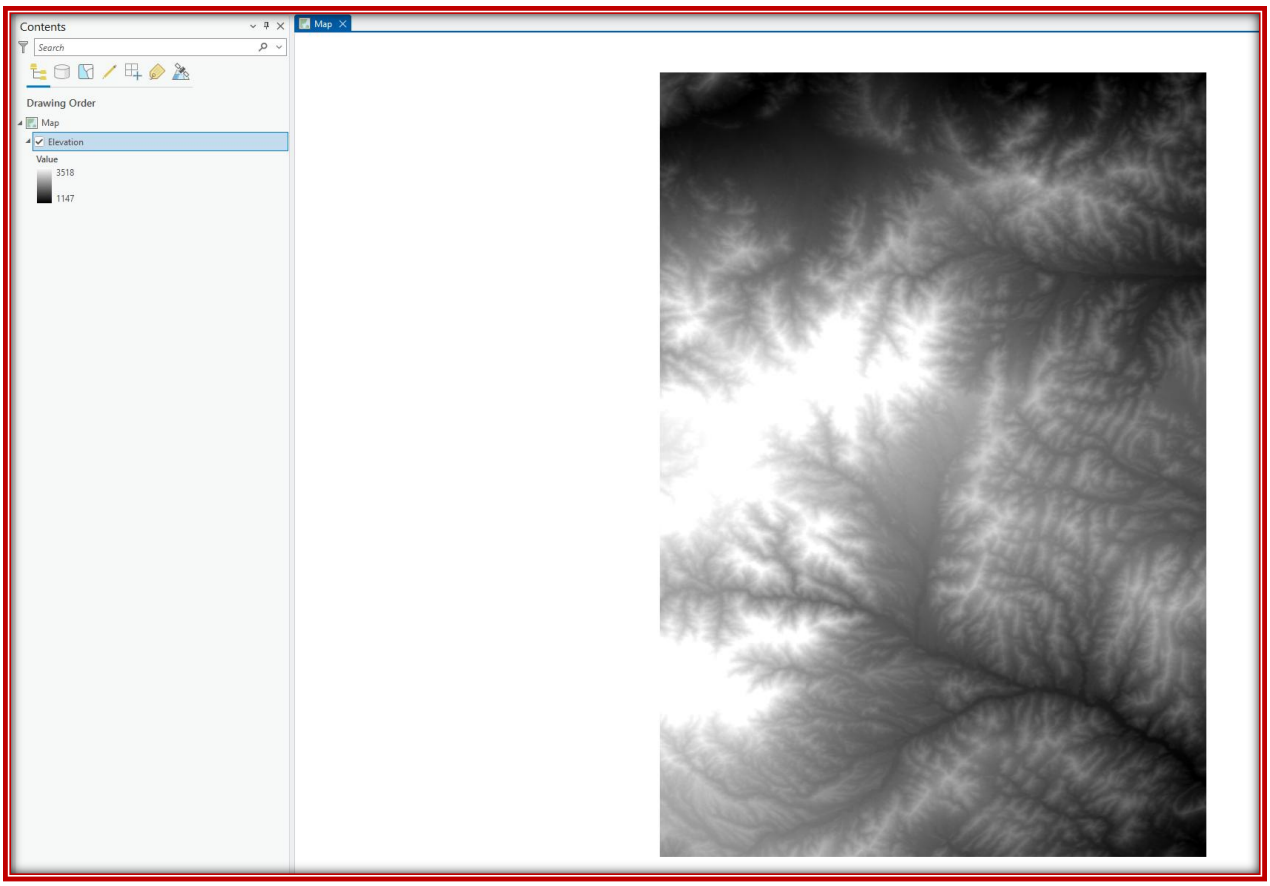
Why It Matters

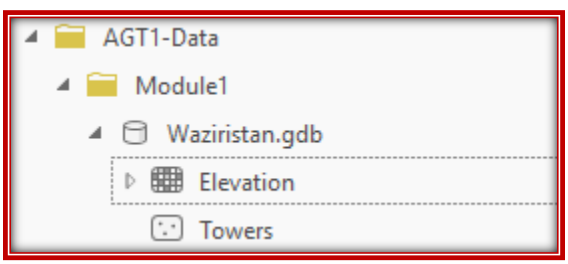
In your engineering and geospatial work, aspect is vital for several practical applications:

- **Hydrology:** Determining which way water will flow off a ridge or mountain.
- **Agriculture & Vegetation:** South-facing slopes in the northern hemisphere receive more sunlight, affecting soil moisture and the types of plants that can grow there.
- **Tactical Analysis:** Identifying "dead space" or areas hidden from an observer's line of sight based on the direction the terrain is sloping.
- **Energy:** Finding the best locations for solar panels, which require specific orientations to maximize efficiency.



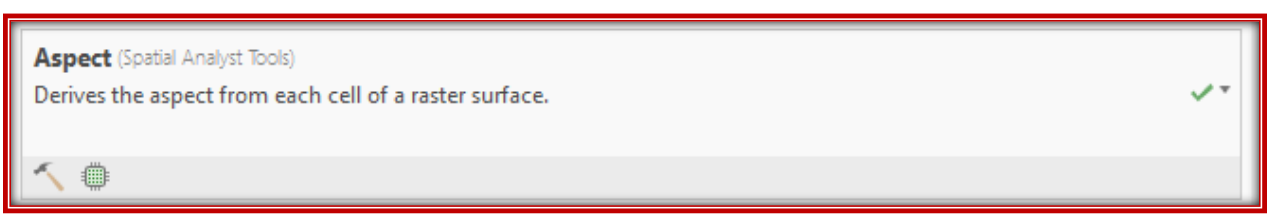
Goal: Calculate aspect of our DEM.

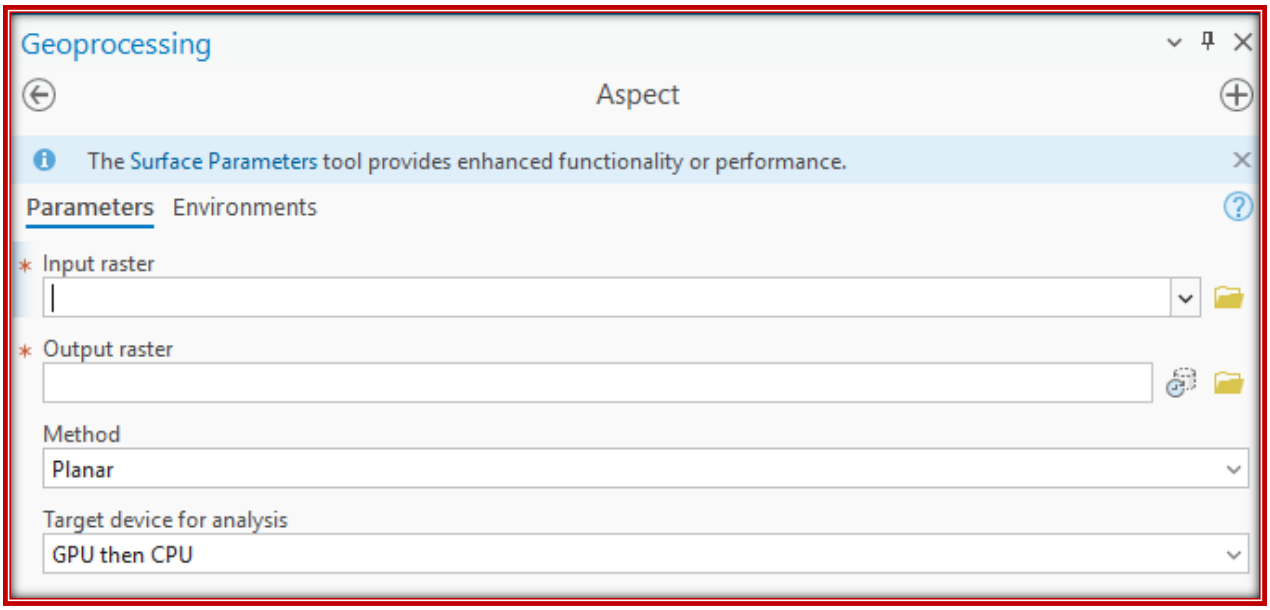




Load Elevation raster (DEM) from Waziristan.gdb

Open **Aspect** tool (spatial analyst):





Parameters

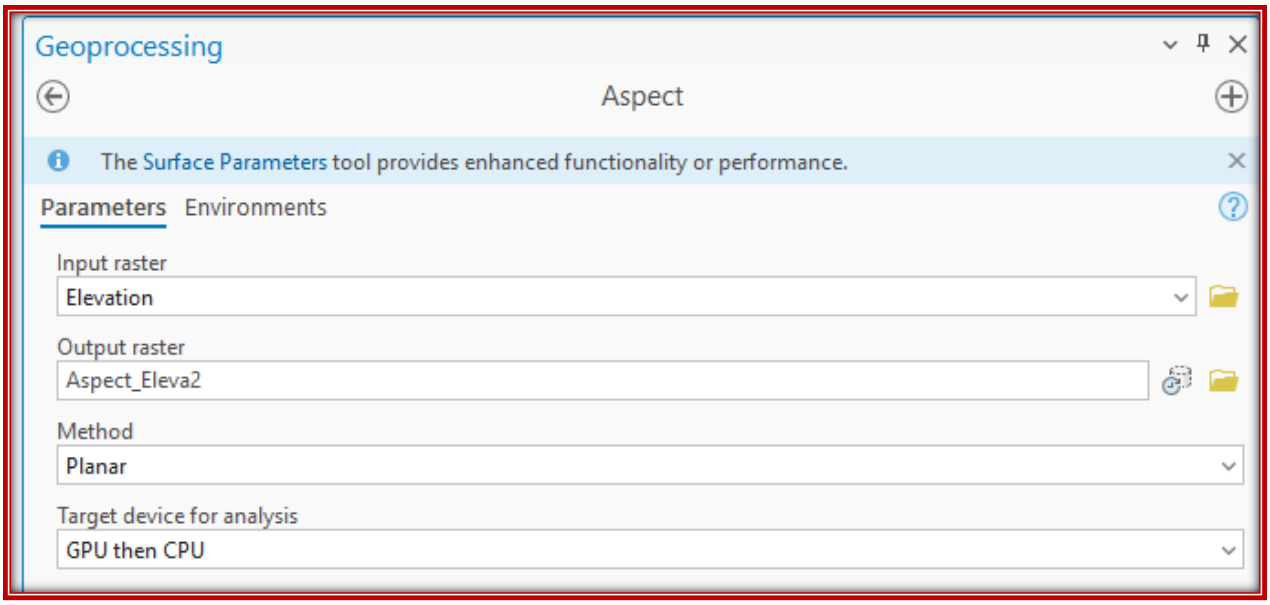
- **Input raster**: This is your source elevation data, typically a Digital Elevation Model (DEM), used to determine the surface direction.
- **Output raster**: The name and location of the new dataset. The output values will range from 0 to 360 (degrees), with -1 used to represent flat areas.
- **Method**: This determines how the tool calculates the direction across the Earth's surface:

    - **Planar**: Calculates direction on a flat 2D plane. It is faster and works well for small, local areas.
    - **Geodesic**: Calculates direction by accounting for the Earth's actual curved shape (ellipsoid). This is the more accurate choice for large-scale analysis or data in geographic coordinates.

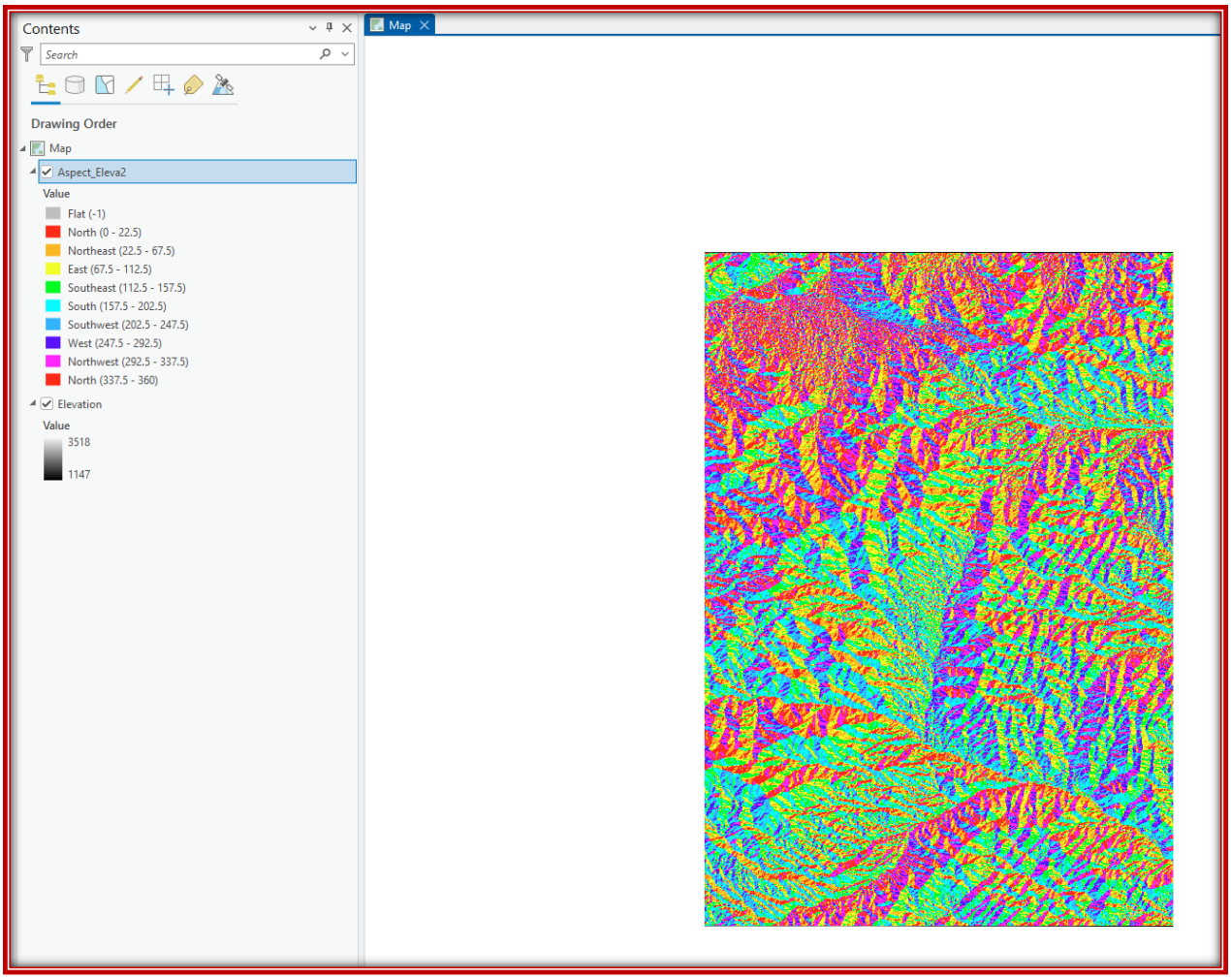
- **Target device for analysis**: Allows you to choose the hardware used for the computation:

    - **GPU then CPU**: Attempts to use your graphics card for faster processing, falling back to the processor if necessary.
    - **CPU only**: Uses only the system's central processor.

Configure as follows:


Geoprocessing
Aspect
The Surface Parameters tool provides enhanced functionality or performance.
Parameters
Environments
Input raster
Elevation
Output raster
Aspect_Eleva2
Method
Planar
Target device for analysis
GPU then CPU

Hit Run



Note that each color indicates a direction as shown in the legend.

Calculate Viewshed

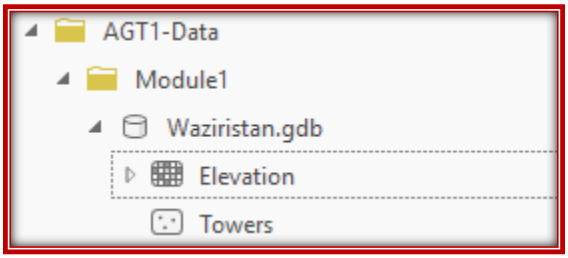
In GIS, **Viewshed** analysis is a geoprocessing operation used to determine the "line-of-sight" from a specific point or set of points across a landscape. It identifies all areas on a surface raster that are visible from an observer's location, as well as which areas are hidden (dead space) by intervening terrain or obstacles.

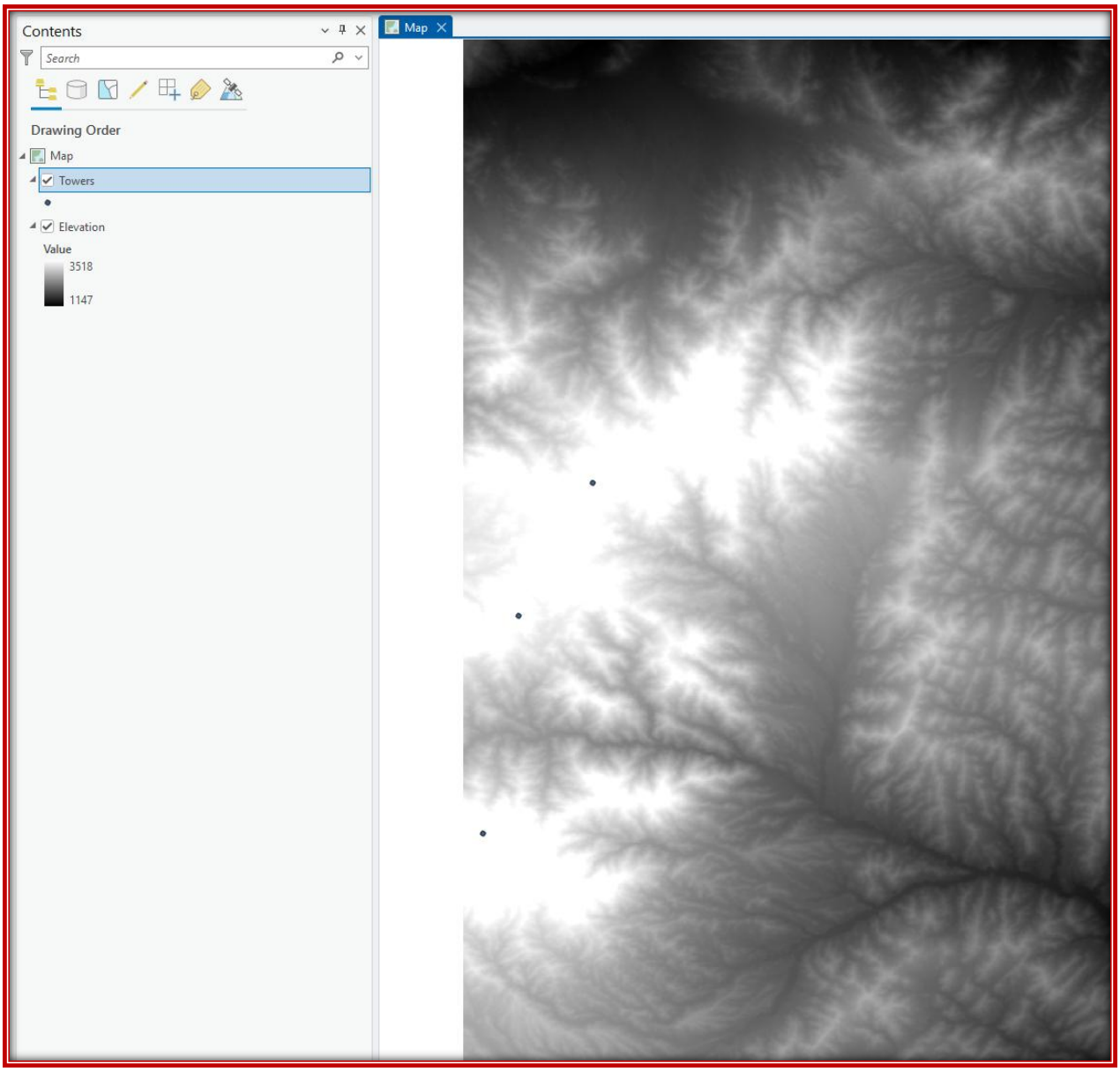
How it Works

The tool calculates visibility by drawing a series of rays from the observer's location to every other cell in the raster. If the elevation of any cell between the observer and the target is higher than the line-of-sight, the target cell is marked as invisible.

When a Viewshed analysis is performed with **multiple observers** (e.g., several point locations in a single feature class), the tool calculates the visibility for each observer and then merges the results into a single output raster.

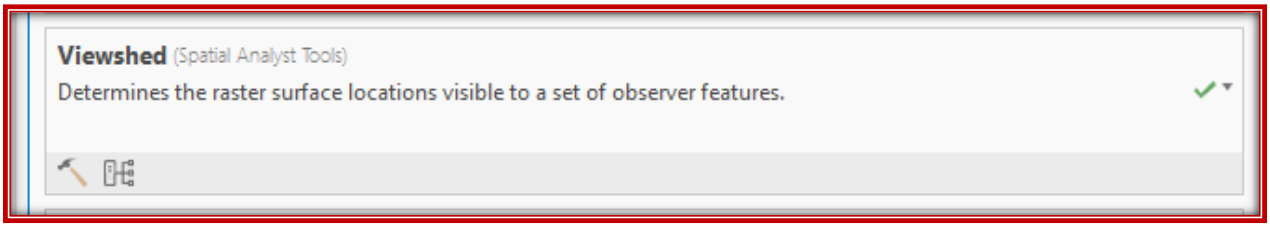
Goal: Calculate Viewshed from our towers of our DEM.





Load Elevation raster (DEM) and Tower.shp from Waziristan.gdb

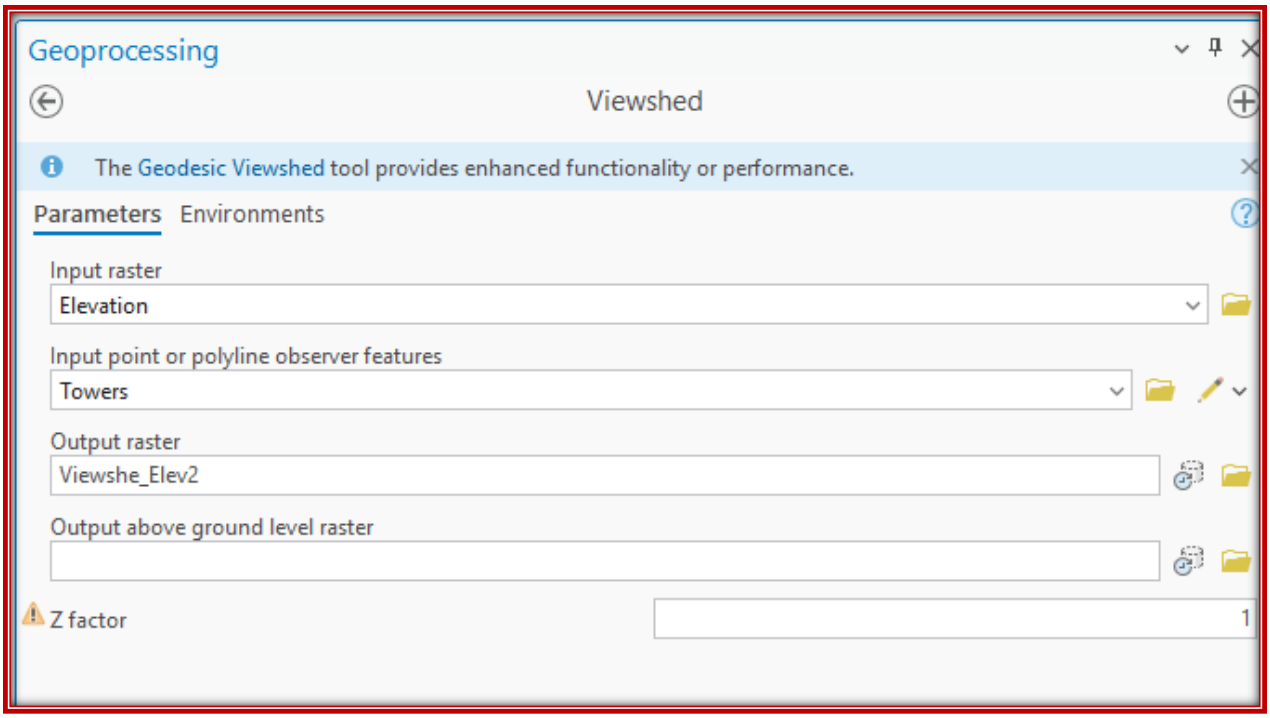
Open **Viewshed** tool (spatial analyst):



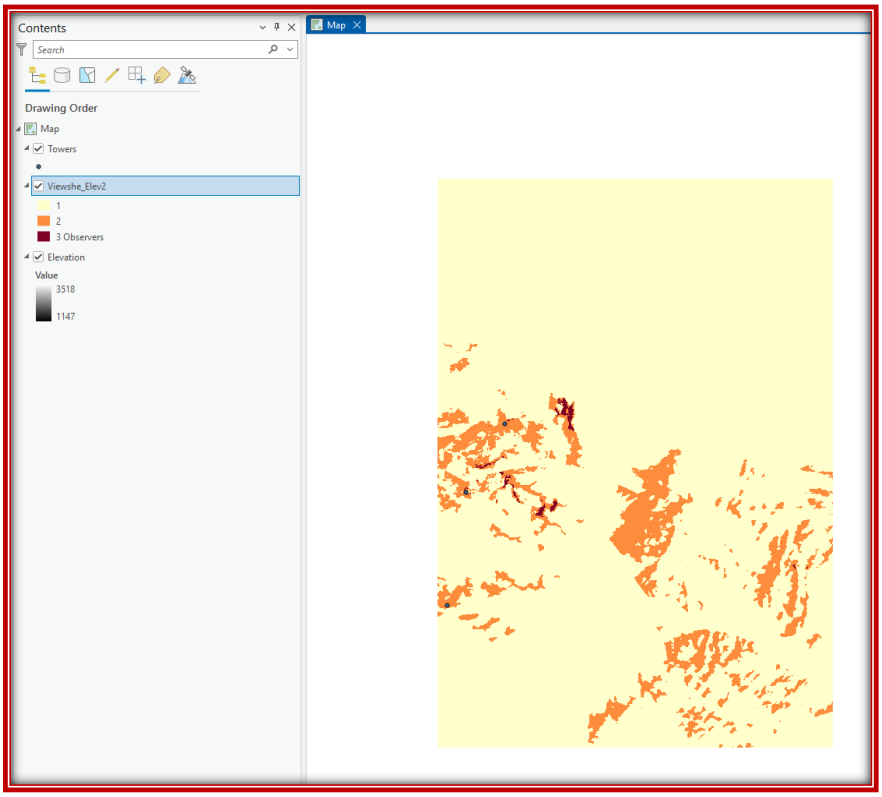
Parameters

- **Input raster**: The elevation surface (typically a DEM) that the tool will use to determine line-of-sight. This surface provides the height values for the terrain.
- **Input point or polyline observer features**: The features that identify the observers.
    - **Points**: Specific locations like a camera, a lookout post, or a person.
    - **Polylines**: Represents an observer moving along a path (e.g., a road or a perimeter fence).
- **Output raster**: The result of the analysis. For each cell in the output, the value represents the number of observer points that can see that specific location. Cells that cannot be seen by any observer are given a value of 0.
- **Output above ground level (AGL) raster**: This is an optional but highly useful output. It creates a raster where each cell value represents the **minimum height** that must be added to that cell to make it visible to at least one observer.
    - *Example:* If a cell has an AGL value of 10, it means that location is currently hidden, but a 10-meter tall tower placed there would be visible to the observer.
- **Z factor**: This is used to adjust for unit differences between the horizontal (X,Y) and vertical (Z) measurements.
    - As with the **Slope** tool, if your map units are in meters but your elevation is in feet, you would enter **0.3048**.
    - Ensuring your **Environment** settings are set to a **Projected Coordinate System (PCS)** is critical here to ensure the X, Y, and Z units are consistent, preventing the "dark result" or mathematical errors caused by decimal degrees.

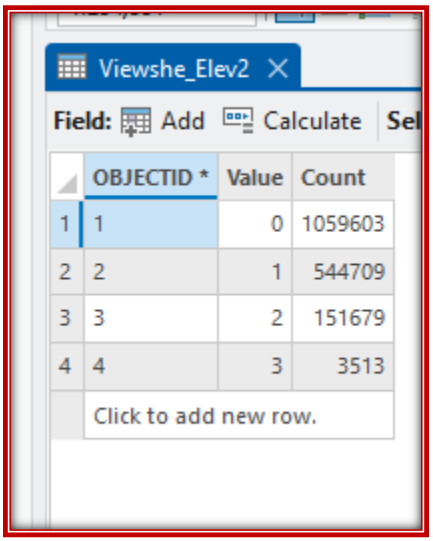
Configure as follows:


Geoprocessing
Viewshed
The Geodesic Viewshed tool provides enhanced functionality or performance.
Parameters
Environments
Input raster
Elevation
Input point or polyline observer features
Towers
Output raster
Viewshe_Elev2
Output above ground level raster
Z factor
1

Hit **Run:**


Contents
Search
Drawing Order
Map
Towers
Viewshe_Elev2
1
2
3 Observers
Elevation
Value
3518
1147
Map

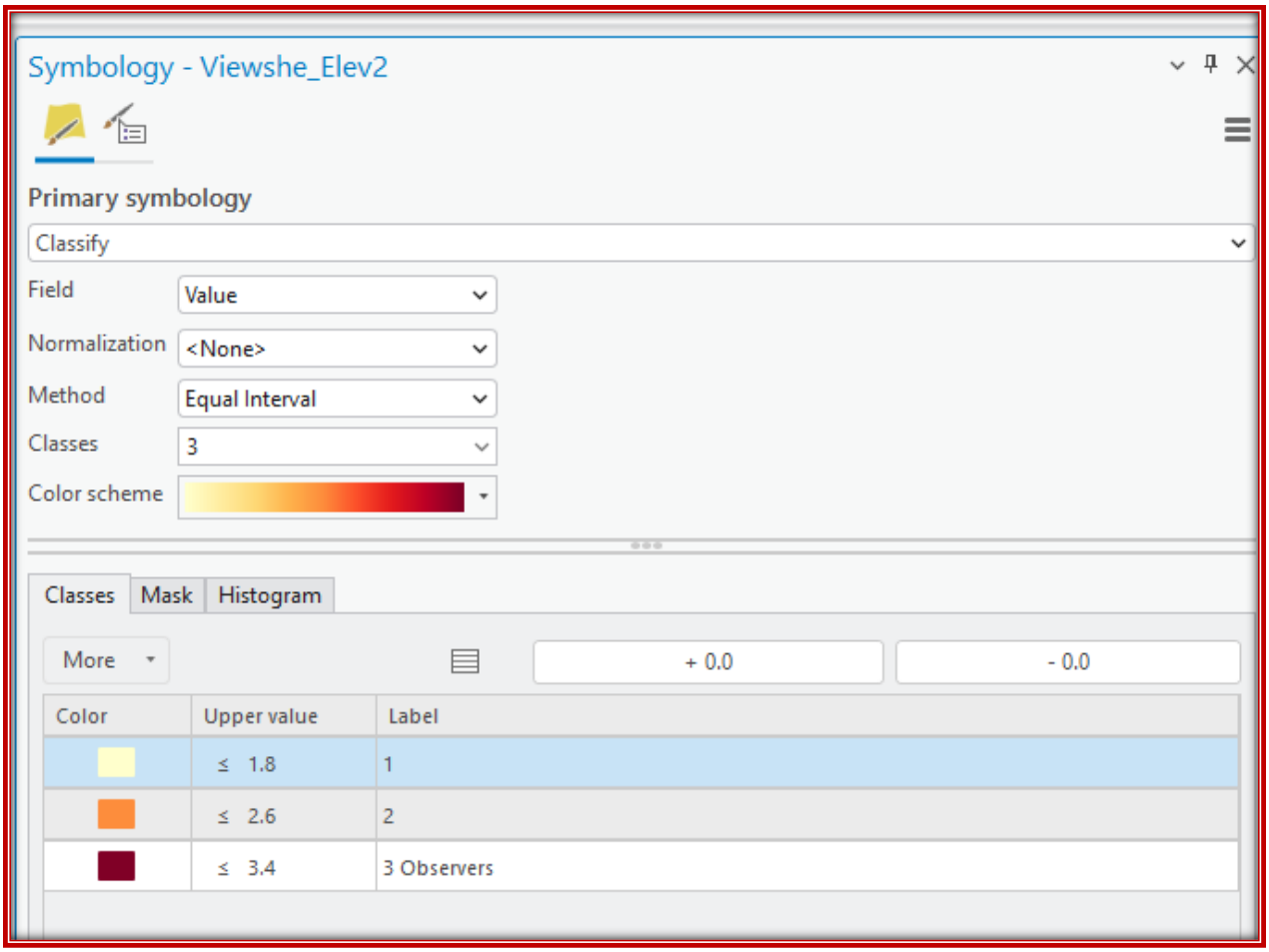
A viewshed has been generated but let's analyze it, Open attributes:



	OBJECTID *	Value	Count
1	1	0	1059603
2	2	1	544709
3	3	2	151679
4	4	3	3513

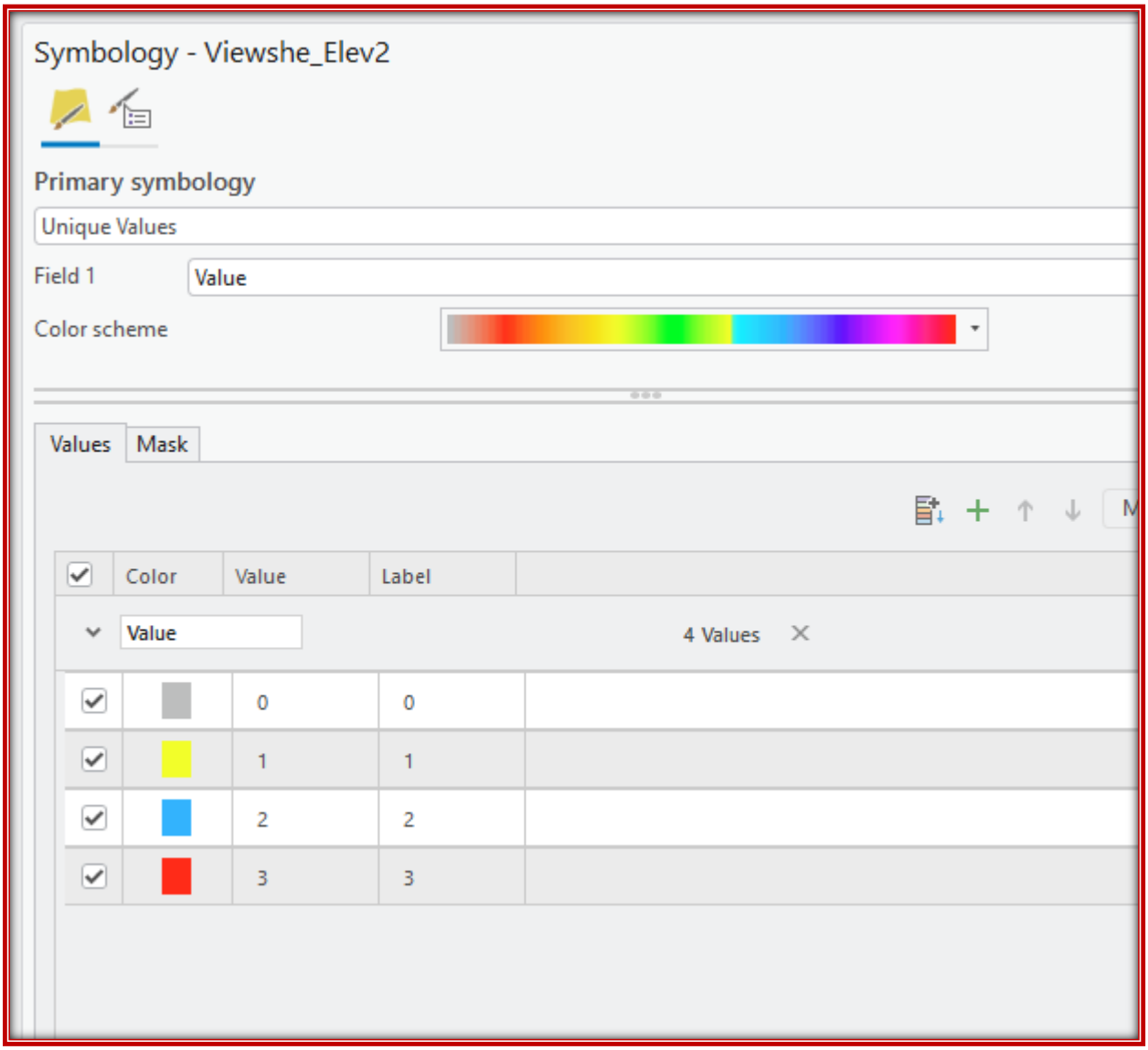
Notice that each cell has a value ranging from **0-3.** This shows the number of observers (towers) that this cell is visible from.

Now let's open symbology for the generated layer:

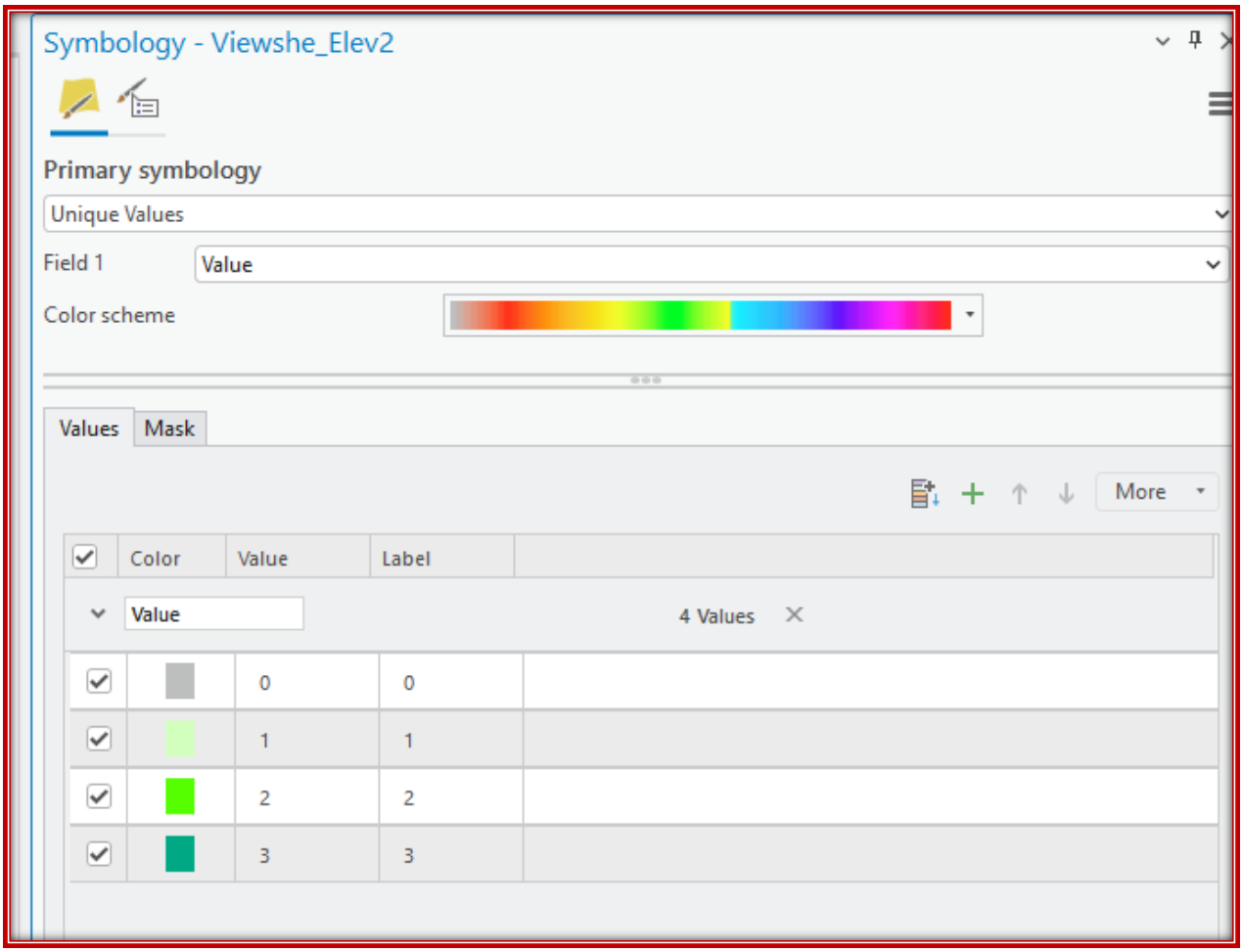


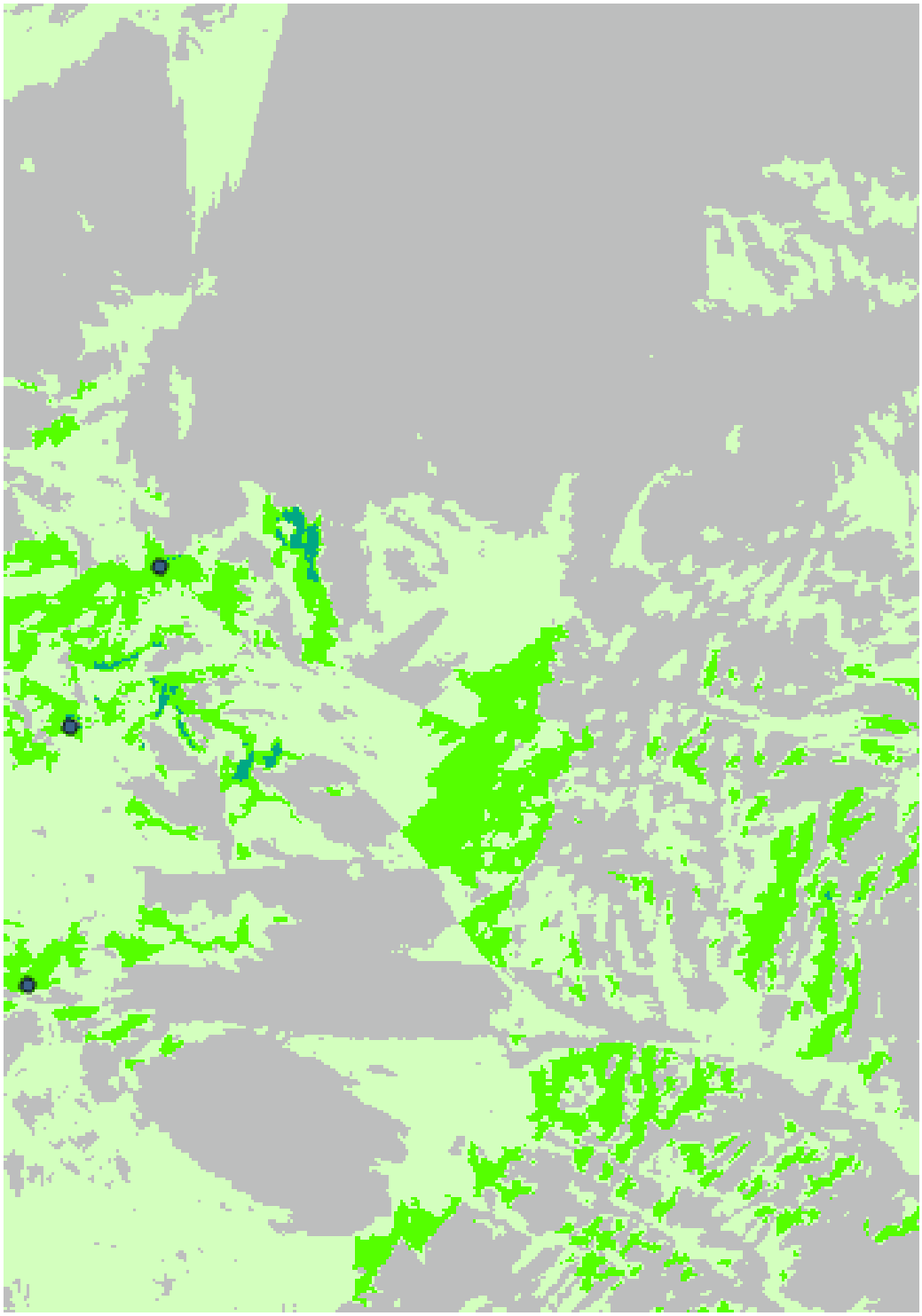
It is assigning a range for classifying the values. This does NOT suit our requirement. Change it to unique values:

`



We ca then assign a different shade of the same color to better visualize it:


Symbology - Viewshe_Elev2
Primary symbology
Unique Values
Field 1
Value
Color scheme
Values
Mask
More
Color
Value
Label
Value
4 Values
0
0
1
1
2
2
3
3



Deliverables (To be submitted)

- Map showing contours with index contours highlighted using **both** methods (Workspace screenshot not layout)
- Map showing hillshade for Waziristan DEM (Workspace screenshot not layout) Map showing slope raster for Waziristan DEM (Workspace screenshot not layout)
- Map showing aspect raster for Waziristan DEM (Workspace screenshot not layout)
- Map showing viewshed for given tower locations using Waziristan DEM (Workspace screenshot not layout)